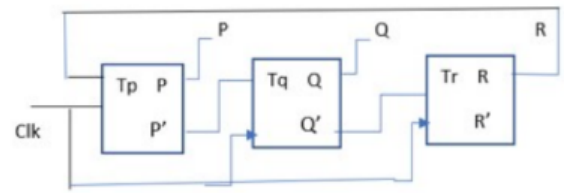


- Q1** Which of the following is NOT logically equivalent to $P \rightarrow Q$?
- (A) $\sim P \vee Q$ (B) $\sim Q \rightarrow \sim P$
 (C) $\sim(P \wedge \sim Q)$ (D) $\sim Q \rightarrow P$
- Q2** Seven people enter a lift. The lift stops only at three floors (which are unspecified). At each of the three floors, no one enters the lift, but at least one person leaves the lift. After the three floor stops, the lift is empty. In how many ways this can happen?
- (A) 1844 (B) 2187
 (C) 1806 (D) 343
- Q3** The coefficients of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$ is
- (A) $C(7, 3) * C(5, 2)$ (B) $C(7, 2) * C(5, 3)$
 (C) $C(7, 3) * C(7, 2)$ (D) $C(5, 3) * C(5, 2)$
- Q4** What is the minimum number of students in a discrete mathematics class to be sure that at least six will receive the same grade, if there are seven possible grades: S, A, B, C, D, P and F?
- (A) 7 (B) 13
 (C) 36 (D) 14
- Q5** In 16-bit 2's Complement representation, the decimal number 28 is
- (A) 1111 1111 0001 1100
 (B) 1111 1111 1111 0100
 (C) 1111 1111 1110 0100
 (D) 1111 1111 1110 0011
- Q6** Simplify the following using K-Map.
 $F(A, B, C, D) = \sum(3, 4, 5, 6, 7, 11, 15)$
- (A) $AB + C'D$ (B) $A'B + CD$
 (C) $AC + BD$ (D) $A'B + B'C$
- Q7** Consider a 3-bit counter, designed using T flip flops as shown below:



Assume the initial state of the counter given by PQR as 000, what are the next three states?

- (A) 011, 101, 000 (B) 001, 010, 111
 (C) 011, 101, 111 (D) 001, 010, 000
- Q8** Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R1 + R2$, which one of the following values of R1 and R2 gives an arithmetic overflow?
- (A) $R1 = 1011$ and $R2 = 1110$
 (B) $R1 = 1100$ and $R2 = 1010$
 (C) $R1 = 0011$ and $R2 = 0100$
 (D) $R1 = 1001$ and $R2 = 1111$
- Q9** In the context of Binding which one of the following is incorrect?
- (A) Static binding is done at Compile Time
 (B) Absolute addresses are known at Load Time
 (C) Internal functions are known at link time
 (D) Target of a function is known at runtime
- Q10** Given
- ```
int arr[3][4][2]
= {{{2, 4}, {7, 8}, {3, 4}, {5, 6}},
 {{7, 6}, {3, 4}, {5, 3}, {2, 3}},
 {{8, 9}, {7, 2}, {3, 4}, {5, 1}}};
```
- Which one of the following can be used to refer to element 1 in the above 3-d array?
- (A)  $\text{arr}[3][4][2]$   
 (B)  $*( * (\text{arr} + 2) + 3) + 1$   
 (C)  $*(\text{arr} + 2) + *(\text{arr} + 3) + *(\text{arr} + 1)$   
 (D)  $\text{arr}[2][3][2]$
- Q11** Allowing different derived classes to be used interchangeably through the interface provided by a common base class is called \_\_\_\_\_
- (A) Derived polymorphism  
 (B) Compile time polymorphism



- (C) Static polymorphism  
(D) Run time polymorphism

**Q12** In \_\_\_\_\_ curve the last and the first vertices of the polygon lie on the curve and all other vertices define the derivatives, order and shape of the curve.

- (A) B-Spline Curve (B) Boxline curve  
(C) Bezier curve (D) Gaussian curve

**Q13** A \_\_\_\_\_ operation combines the rows of the two relations and outputs a new relation that has both input relations' rows in it.

- (A) difference  
(B) union  
(C) cartesian product  
(D) project

**Q14** Consider the relation  $R(P, Q, R, S, T, U, V, W, X, Y)$  and FDs:

$$\{P, Q, S\} \rightarrow T$$

$$R \rightarrow Y$$

$$\{P, Q\} \rightarrow V$$

$$\{R, X\} \rightarrow X$$

$$Q \rightarrow U$$

$$V \rightarrow \{W, X\}$$

Which of the following is a candidate key?

- (A)  $\{P, Q, R, S\}$   
(B)  $\{R, Q, X\}$   
(C)  $\{V, X\}$   
(D) None of the above

**Q15** Which RAID level is appropriate for storing the log file of a ticket booking system where a huge database update is required?

- (A) Level 0 (B) Level 1  
(C) Level 2 (D) Level 3

**Q16** If we convert a decision tree to a set of logical rules, then:

- (A) The internal nodes in a branch are connected by AND and branches by AND  
(B) The internal nodes in a branch are connected by OR and branches by OR  
(C)

The internal nodes in a branch are connected by AND and branches by OR

- (D) The internal nodes in a branch are connected by OR and branches by AND

**Q17** Model of protection can be viewed as

- (A) Owner matrix (B) Domain matrix  
(C) Access Matrix (D) Global matrix

**Q18** Process in memory also contain the heap section. What is the use of heap section?

- (A) Contains temporary data  
(B) Dynamically allocate the memory during process run time  
(C) Contains Process priority and pointers to scheduling queues  
(D) Contains amount of CPU and real time use

**Q19** How many page faults occur for optimal algorithm for the following reference string, with four frames?

1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8

- (A) 8 (B) 7  
(C) 6 (D) 5

**Q20** Degree of concurrency in thread can be arranged in the manner.

- (A) one-to-one > many-to-one > many-to-many  
(B) one-to-one > many-to-many > many-to-one  
(C) many-to-many > many-to-one > one-to-one  
(D) None of the above

**Q21** Which of the following is not one of the principles of good coding?

- (A) Create unit tests before you begin coding.  
(B) Create visual layout that aids understanding.  
(C) Refactor the code after you completes the first coding pass.  
(D) Write self-documenting code not program documentation.

**Q22** If P is the risk probability, L is the loss, then Risk exposure (RE) is computed as \_\_\_\_\_

- (A)  $RE = P / L$   
(B)  $RE = P + L$   
(C)  $RE = P * L$   
(D)  $RE = 2 + (P * L)$



- Q23** Which of the following is a technique used for component wrapping?  
 (A) Black-box wrapping  
 (B) Clear-box wrapping  
 (C) Gray-box wrapping  
 (D) White-box wrapping
- Q24** The cyclomatic complexity metric provides the designer with information regarding the number of \_\_\_\_  
 (A) Cycles in the program  
 (B) Errors in the program  
 (C) Independent logic paths in the program  
 (D) Statements in the program
- Q25** In a \_\_\_\_\_ the left and right subtrees for any given node, differ in height by more than one.  
 (A) Balanced Binary Tree  
 (B) Weight balanced binary tree  
 (C) Height balanced tree  
 (D) Unbalanced Binary Tree
- Q26** FFTs are \_\_\_\_\_ techniques used to encode digital video and audio information and helps analyse the signal in \_\_\_\_\_ domain.  
 (A) Encryption, time  
 (B) Encryption, frequency  
 (C) Compression, time  
 (D) Compression, frequency
- Q27** Given six weights 4,15,25,5,8,16 construct a  $2 - T$  tree. What is the minimum weighted path length?  
 (A) 231 (B) 172  
 (C) 73 (D) 77
- Q28** Suppose we store  $n$  keys in a hash table of size  $m = n^2$  using a hash function  $h$  randomly chosen from a universal class of hash functions. Then the probability is \_\_\_\_\_ that there are any collisions.  
 (A) greater than  $1/5$   
 (B) greater than  $1/2$   
 (C) less than  $1/2$   
 (D) less than  $1/5$

Q29

In Pumping Lemma for context-free languages, to say a language is satisfying pumping lemma, what is the minimum length of ' $vx$ ' together, if you consider the string as ' $uvwxy$ '?

- (A) 0 (B) 1  
 (C) 2 (D)  $n$

- Q30** Which of the following is  $\varepsilon$ -free grammar?  
 (A) Type-0 (B) Type-1  
 (C) Type-2 (D) Type-3
- Q31** While processing a string by an LR parser, the reading of the given string by the parser is-  
 (A) From right to left  
 (B) From left to right  
 (C) Can be either from left to right or right to left  
 (D) From the centre of the given string
- Q32** What is the minimum and maximum number of initial, dead, unreachable and final states in a valid ' $n$ ' state finite automaton, where the language accepted is not ' $\varphi$ '?  
 (A) Minimum: 1, 1, 1, 1; Maximum: 1,  $(n - 1)$ ,  $(n - 1)$ ,  $n$   
 (B) Minimum: 1, 0, 0, 0; Maximum: 1,  $(n - 1)$ ,  $(n - 1)$ ,  $n$   
 (C) Minimum: 1, 0, 0,  $n$ ; Maximum: 1,  $n$ ,  $(n - 1)$ ,  $(n - 1)$   
 (D) Minimum: 1, 0, 0, 1; Maximum: 1,  $(n - 1)$ ,  $(n - 1)$ ,  $n$
- Q33** Which protocol is used to acquire the MAC Address of a host whose IP address is known?  
 (A) Dynamic Host control protocol  
 (B) Address Resolution Protocol  
 (C) Reverse Address Resolution Protocol  
 (D) User Datagram Protocol
- Q34** In CRC coding if the data word is 111111, divisor is 1010 and the remainder is 110. Which of the following code is true?  
 (A) 011111101 (B) 001010110  
 (C) 111111110 (D) 110111111
- Q35** In Go-back-N protocol, the values of  $S_f$  (Send window, the first outstanding packet) = 3,  $S_n$  (Send window, the next packet to be send) = 7,



after receiving the packet with AckNo.6, the value of  $S_n$  is \_\_\_\_

- (A) 6 (B) 7  
(C) 5 (D) 4

**Q36** What is the minimum Hamming Distance in a block code if we want to be able to detect up to (S) errors?

- (A) S (B) S + 1  
(C) 1 (D) 0

**Q37** Which search technique is complete and optimal, when  $h(n)$  is consistent?

- (A) Best First Search  
(B) Depth First Search  
(C)  $A^*$  Search  
(D) Breadth First Search

**Q38** What is the process of capturing inference process as a single inference rule?

- (A) Ponens  
(B) Clauses  
(C) Generalized modus ponens  
(D) Variables

**Q39** Fuzzy logic is a form of \_\_\_\_

- (A) Two valued logic  
(B) Crisp set logic  
(C) Binary set logic  
(D) Many valued logic

**Q40** The 4 Ws problem canvas helps in identifying the key elements related to the problem. 4 Ws problem canvas is a part of \_\_\_\_

- (A) Problem scoping  
(B) Data acquisition  
(C) Data modeling  
(D) Problem Evaluation

**Q41** Consider the following statements:

- A. A group can have zero elements.  
B. A group cannot have any element which is idempotent except the identity element.  
C. An isomorphism from a group  $\langle G, * \rangle$  to  $\langle G, * \rangle$  is called an automorphism.  
D. If every element in a group is its own inverse, then the group must be abelian.

Choose the correct answer about the four statements given above.

- (A) B, C and D are correct  
(B) Only A and C are correct  
(C) Only B and C are correct  
(D) All A, B, C and D are correct.

**Q42** Consider the following statements:

- A. The number of labelled binary trees possible with 'n' nodes is  $(2n)! / ((n+1)! n!)$   
B. The number of binary search trees possible with 'n' nodes is  $(2n)! / ((n+1)! n!)$   
C. The number of unlabeled binary trees possible with 'n' nodes is  $C(2n, n) / (n+1)!$   
D. The number of simple, undirected and labelled graphs possible with 'n' vertices is  $(2^{n(n-1)/2})$

Choose the correct answer from the options given below:

- (A) (A, B) and (C) are only correct  
(B) (A, B) and (D) are only correct  
(C) (B, C) and (D) are only correct  
(D) (A, C) and (D) are only correct

**Q43** The relation

$$R = \{(1, 3), (1, 1), (3, 1), (1, 2), (3, 3), (4, 4)\}$$

defined in  $\{1, 2, 3, 4\}$  is \_\_\_\_

- (A) Reflexive, Symmetric but not Transitive  
(B) Symmetric, anti-symmetric and transitive  
(C) Asymmetric, anti-symmetric and not transitive  
(D) Not reflexive, not symmetric and not transitive

**Q44** Consider the Boolean expression given by

$$F = (X + Y + Z) (X' + Y) (Y' + Z).$$

Which of the following Boolean expressions is/are equivalent to  $F'$  (Complement of F)?

- A.  $(X' + Y' + Z') (X + Y') (Y + Z')$   
B.  $(XY' + Z')$   
C.  $(X + Z') (Y' + Z')$   
D.  $(XY' + YZ' + X'Y'Z')$

Choose the correct answer from the options given below:

- (A) A, B and C Only  
(B) B, C and D Only  
(C) A, C and D Only



(D) A, B and D Only

**Q45** Which of the following statement is/are True?

- A. The processor has direct access to both primary and secondary memory
- B. Primary memory stores the active instructions and data for the program being executed on the process
- C. Secondary memory is used as a backup memory
- D. Memory system is implemented on a single level memory

Choose the correct answer from the options given below:

- (A) A and B only
- (B) B and C only
- (C) A, B and C only
- (D) A and D Only

**Q46** Inheritance provides the benefits of

- A. shared interfaces
- B. run-time polymorphism
- C. less redundant information
- D. dynamic binding

Choose the Correct answer from the options given below:

- (A) A, B, C only
- (B) A, B, D only
- (C) B, C, D only
- (D) A, C, D only

**Q47** Which of the following is correct using the constructor?

- A. string s1;
- B. string s2 (s1);
- C. string s3{7};
- D. string s4{'a'};
- E. string s5{7, 'a'};

Choose the most appropriate answer from the options given below:

- (A) A, B, D, E only
- (B) A, B, C only
- (C) A, B, D only
- (D) A and B only

**Q48** Suppose that a corporate website appearance varies according to the department that owns various documents. All the documents need to

follow the corporate look and feel. The HR department does not need a separate style sheet. It needs only the sheet containing the difference from the corporate sheet.

Which of the following statements justify the above scenario?

- A. The styles sheets cannot stack on each other
- B. The style sheets can stack on each other
- C. The style sheets can override each other
- D. The style sheets cannot override each other

Choose the most appropriate answer from the options given below:

- (A) A, C Only
- (B) B, C Only
- (C) B, D only
- (D) A, D only

**Q49** Let  $T_1$  be a transaction that transfer Rs. 50 from account A to account B. This transaction can be defined as follows:

| Step | Operation     |
|------|---------------|
| 1    | Read(A)       |
| 2    | $A := A - 50$ |
| 3    | Write(A)      |
| 4    | Read(B)       |
| 5    | $B := B + 50$ |
| 6    | Write(B)      |

What will be the status of a database just after 3<sup>rd</sup> step but before 5<sup>th</sup> step of transaction execution?

- (A) Always consistent
- (B) May be temporarily inconsistent
- (C) Must be abort
- (D) Always rollback

**Q50** Consider the relation Student(stuid, dept, clsroll)

The following functional dependencies are given:

- FD1: stuid  $\rightarrow$  dept
- FD2: dept  $\rightarrow$  stuid
- FD3: stuid  $\rightarrow$  clsroll
- FD4: clsroll  $\rightarrow$  stuid
- FD5: dept  $\rightarrow$  clsroll

Which functional dependencies should be removed to obtain the canonical cover of the set?



- A. FD2 and FD3
- B. FD1 and FD5
- C. FD5
- D. FD1

Choose the correct answer from the options given below:

- (A) A, B only
- (B) A, C only
- (C) B, C only
- (D) A, B, C only

**Q51** Which of the following statements are true?

A. If the time quantum is extremely small, round robin can result in a large number of context switches.

B. The average turnaround time of a set of processes does not necessarily improve as the time-quantum size increases.

C. The average turnaround time can be improved if most processes finish their next CPU burst in a single time quantum.

Choose the correct answer from the options given below:

- (A) A only
- (B) A, B only
- (C) B, C only
- (D) A, B, C

**Q52** Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds.

| Process Name | Arrival Time | Execution Time |
|--------------|--------------|----------------|
| A            | 0            | 6              |
| B            | 3            | 2              |
| C            | 5            | 4              |
| D            | 7            | 6              |
| E            | 10           | 3              |

Using the shortest remaining time first scheduling algorithm, the average process turnaround (in milliseconds) is \_\_\_\_

- (A) 7.2
- (B) 8.2
- (C) 6
- (D) 9

**Q53** Select the correct order of basic page replacement.

- A. Page replacement algorithm is used in case there is no free frame
- B. The process is continued by restarting the instruction caused by TRAP

- C. Location of desired frame is found from disk
- D. Desired page is brought into newly freed frame and page & frame tables are updated

- (A) A, B, D, C
- (B) C, D, A, B
- (C) C, A, D, B
- (D) A, D, B, C

**Q54** Given below are two statements:

Statement I: Software testing must be recognized as a significant effort, and its planning must begin at the earliest possible time.

Statement II: The cost of software testing is an important contribution in determining overall software development costs.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true.
- (B) Both Statement I and Statement II are false.
- (C) Statement I is true but Statement II is false.
- (D) Statement I is false but Statement II is true.

**Q55** Which of the following need not be assessed during unit testing?

- A. Algorithmic performance.
- B. Number of lines of code.
- C. Error handling.
- D. Execution paths.

Choose the correct answer from the options given below:

- (A) A, B, C only
- (B) A, B, D only
- (C) A, C, D only
- (D) B, C, D only

**Q56** Which of the following rotations are correctly explained in the context of an AVL tree.

A. LL rotation: The new node is inserted in the left sub-tree of the left sub-tree of the critical node.

B. RR rotation: The new node is inserted in the right sub-tree of the right sub-tree of the critical node.

C. LR rotation: The new node is inserted in the right sub-tree of the left sub-tree of the critical node.

D. RL rotation: The new node is inserted in the left sub-tree of the right sub-tree of the critical node



Choose the correct answer from the options given below:

- (A) A, D only (B) A, C Only  
(C) A, B, C and D (D) B, C Only

- Q57** In the context of a Hamiltonian problem, which of the following are correct?  
A. Hamiltonian cycle problem is NP-Complete  
B. If  $G$  is an undirected Bipartite graph with an odd number of vertices, then  $G$  is Hamiltonian.  
C. Hamiltonian path problem cannot be solved in polynomial time on directed acyclic graph  
D. If  $G$  is a connected undirected graph with at least three vertices and  $G^3$  is the graph obtained by connecting all pairs of vertices that are connected by a path in  $G$  of length at most 3.  
E. If any NP complete problem is polynomial time solvable then  $(P=NP)$ .

Choose the most appropriate answer from the options given below:

- (A) A, B, C only (B) B, D, E only  
(C) A, D, E only (D) A, C, E only

- Q58** For matrix multiplication, if we use divide and conquer algorithm, then \_\_\_\_  
A. the running time is  $T(n) = \Theta(n^3)$   
B. the running time is  $T(n) = \Theta(n^2 \log n)$   
C. The recurrence can be defined as  $T(n) = 8T(n/2) + \Theta(n^2)$   
D. the recurrence relation is  $T(n) = 8T(2n) + \Theta(n^2 \log n)$   
E. Masters theorem, one of the cases is applicable  
Choose the most appropriate answer from the options given below:

- (A) A, C, E only (B) A, D, E only  
(C) B, D, E only (D) B, C, D only

- Q59** Which of the following Finite Automata, recognizes the language  $L = \{a, b\}^*$  ?  
A.  $M = (\{q_0\}, \{a, b\}, \delta, q_0, \{q_0\})$  where,  $\delta$  is

|       | a         | b         |
|-------|-----------|-----------|
| $q_0$ | $\{q_0\}$ | $\{q_0\}$ |

- B.  $M = (\{q_0, q_1\}, \{a, b\}, \delta, q_0, \{q_0, q_1\})$  where,  $\delta$  is

|       | a              | b              |
|-------|----------------|----------------|
| $q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_1\}$ |
| $q_1$ | $\{q_1\}$      | $\{q_1\}$      |

|       | a              | b              |
|-------|----------------|----------------|
| $q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_1\}$ |
| $q_1$ | $\{q_1\}$      | $\{q_1\}$      |

C.  $M = (\{q_0, q_1\}, \{a, b\}, \delta, q_0, \{q_0, q_1\})$  where,  $\delta$  is

|       | a              | b         |
|-------|----------------|-----------|
| $q_0$ | $\{q_0, q_1\}$ | $\{q_0\}$ |
| $q_1$ | -              | $\{q_1\}$ |

D.  $M = (\{q_0, q_1, q_2\}, \{a, b\}, \delta, q_0, \{q_0\})$  where,  $\delta$  is

|       | a         | b         |
|-------|-----------|-----------|
| $q_0$ | $\{q_0\}$ | $\{q_0\}$ |
| $q_1$ | $\{q_2\}$ | -         |
| $q_2$ | $\{q_0\}$ | $\{q_2\}$ |

- (A) A and B (B) A and D  
(C) A, B and C (D) A, B, C and D

- Q60** Which of the following statement(s) are correct?

- A. Syntax Directed Translation can be applied in Type Checking.  
B. Syntax Directed Translation can be used in 3-address code generation.  
C. Syntax Directed Translation can be used in construction of syntax tree.  
D. Syntax Directed Translation can be used to recognize valid tokens.

Choose the most appropriate answer from the options given below:

- (A) A, B and C (B) A, B, and D  
(C) A, C and D (D) B, C and D

- Q61** Consider a computer network using the distance vector routing algorithm in its network layer. The Objective is to find the shortest cost path from router  $R$  to routers  $P$  and  $Q$ . Assume that  $R$  does not initially know the shortest routes to  $P$  and  $Q$ . Assume that  $R$  has three neighboring routers, namely:  $X$ ,  $Y$  and  $Z$ . During one iteration  $R$  measures its distance as 3, 2, and 5 from its neighbors  $X$ ,  $Y$  and  $Z$ , respectively. Router  $R$  getting routing vectors from its neighbors that indicates a distance of 7, 6 and 5 from router  $X$ ,  $Y$  and  $Z$  to router  $P$ , respectively. The routing vector also indicates that the distance to router  $Q$  from Routers  $X$ ,  $Y$  and  $Z$  are 4, 6, 8



respectively. Which of the following statements is/are True with respect to the new routing table at router  $R$  after updation during this iteration.

- A. The distance from  $R$  to  $P$  will be stored as 10
- B. The distance from  $R$  to  $Q$  will be stored as 7
- C. The next hop router for a packet from  $R$  to  $P$  is  $Y$
- D. The next hop router for a packet from  $R$  to  $Q$  is  $Z$

Choose the correct answer from the options given below:

- (A) B and C only                      (B) A and C only
- (C) C and D only                      (D) B and D only

**Q62** Consider the routing table of an organization's router as shown in the table given below

| Subnet Number | Subnet Mask   | Next Hop    |
|---------------|---------------|-------------|
| 12.20.164.0   | 255.255.252.0 | R1          |
| 12.20.170.0   | 255.255.254.0 | R2          |
| 12.20.168.0   | 255.255.254.0 | Interface 0 |
| 12.20.166.0   | 255.255.254.0 | Interface 1 |
| Default       | -             | R3          |

Which of the following prefixes in CIDR Notation can be collectively used to correctly aggregate all of the subnets in the routing table?

- A. (12.20 .164 .0 / 20)
- B. (12.20 .164 .0 / 22)
- C. (12.20 .164 .0 / 21)
- D. (12.20 .168 .0 / 22)

Choose the correct answer from the options given below:

- (A) A and B Only
- (B) A and C Only
- (C) C and D Only
- (D) B and D Only

**Q63** Consider two hosts  $P$  and  $Q$  are connected through a router ( $R$ ). The maximum transfer unit (MTU) value of the link between ( $P$ ) and ( $R$ ) is 1500 bytes and between ( $R$ ) and ( $Q$ ) is 820 bytes. A TCP segment of size 1400 bytes is transferred from  $P$  to  $Q$  through  $R$  with IP identification value of  $0 \times 1234$ . Assume that IP header size is 20 bytes. Further, the packet is allowed to be fragmented, meaning the Don't

Fragment (DF) flag in the IP Header is not set by  $P$ .

Which of the following statements is/are not true?

- A. Two fragments are created at ( $R$ ) and IP datagram size carrying the second fragment is 620 bytes
- B. If the second fragment is lost then ( $R$ ) resend the fragment with IP Identification value of  $0 \times 1234$
- C. If the second fragment is lost then  $P$  is required to resend the entire TCP segment
- D. TCP destination port can be determined by analyzing the second fragment only

Choose the correct answer from the options given below:

- (A) A and B Only                      (B) A and C Only
- (C) C and D Only                      (D) B and D Only

**Q64** Consider the following features of Neural networks.

- A. It is essentially a machine learning algorithm.
- B. It is useful when solving the problems for which the data set is very large.
- C. They are able to extract features without input from the programmer.
- D. These are systems modeled on the human brain and nervous system.

Choose the most appropriate answer from the options given below:

- (A) A Only
- (B) A & B Only
- (C) A, B & C Only
- (D) A, B, C & D

**Q65** Consider the following draw backs is/are of Rule based approach is/are:

- A. The learning is static.
- B. Any changes made to the original data will not be considered.
- C. Once trained the model cannot improve on the basis of feedback.

Choose the correct answer from the options given below:

- (A) All the above statements are correct



- (B) The statements A & B only are not correct  
 (C) The statements B & C only are not correct  
 (D) All the above statements are not correct

**Q66** Match List I with List II

| List I                           | List II                |
|----------------------------------|------------------------|
| A. Linear Program                | I. Prim's Algorithm    |
| B. Predicate Logic               | II. In order           |
| C. Tree Traversal                | III. Simplex Algorithm |
| D. Minimum Spanning Tree Problem | IV. First order logic  |

Choose the correct answer from the options given below:

- (A) A-IV; B-III; C-I; D-II  
 (B) A-III; B-IV; C-II; D-I  
 (C) A-II; B-IV; C-I; D-III  
 (D) A-III; B-IV; C-I; D-II

**Q67** Match List I with List II

| List I | List II                        |
|--------|--------------------------------|
| A. DAA | I. Program Control Instruction |
| B. LXI | II. Interrupt instruction      |
| C. RST | III. Data Movement Instruction |
| D. JMP | IV. Arithmetic Instruction     |

Choose the correct answer from the options given below:

- (A) A-IV, B-II, C-I, D-III  
 (B) A-IV, B-III, C-II, D-I  
 (C) A-IV, B-II, C-III, D-I  
 (D) A-III, B-IV, C-II, D-I

**Q68** Match List I with List II

| List I                 | List II                   |
|------------------------|---------------------------|
| A. 2-D Array Input     | I. Liquid Crystal Display |
| B. Transmissive device | II. Flat-bed scanner      |
| C. Binary image        | III. Camera               |
| D. 1-D Array input     | IV. Inkjet printer        |

Choose the correct answer from the options given below:

- (A) A-I, B-III, C-II, D-IV  
 (B) A-III, B-I, C-IV, D-II

- (C) A-III, B-I, C-II, D-IV  
 (D) A-I, B-II, C-IV, D-III

**Q69** Match List I with List II

| List I                                      | List II                                                                                                      |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| A. Timestamp protocol                       | I. As long as the transaction is already running, it will try to obtain the required lock in the first part. |
| B. Strict two-phase locking protocol        | II. Indicates final lock of the schedule.                                                                    |
| C. Two-phase locking                        | III. Ensures freedom from deadlock.                                                                          |
| D. Lock point in two-phase locking protocol | IV. Ensures cascade-less rollback                                                                            |

Choose the correct answer from the options given below:

- (A) A-III, B-IV, C-II, D-I  
 (B) A-III, B-IV, C-I, D-II  
 (C) A-IV, B-III, C-II, D-I  
 (D) A-IV, B-II, C-I, D-III

**Q70** Match List I with List II

| List I                             | List II                                                     |
|------------------------------------|-------------------------------------------------------------|
| A. (reference bit=0, modify bit=0) | I. Must write out before replacement.                       |
| B. (reference bit=0, modify bit=1) | II. Probably will be used again soon and need to write out. |
| C. (reference bit=1, modify bit=0) | III. Best page to replace.                                  |
| D. (reference bit=1, modify bit=1) | IV. Probably will be used again soon.                       |

Choose the correct answer from the options given below:

- (A) A-II, B-I, C-III, D-IV  
 (B) A-II, B-I, C-IV, D-III



(C) A-III, B-I, C-II, D-IV

(D) A-III, B-I, C-IV, D-II

**Q71** Match List I with List II

| List I                            | List II                 |
|-----------------------------------|-------------------------|
| A. Condition coverage             | I. Black box testing    |
| B. Equivalence class Partitioning | II. System testing      |
| C. Volume testing                 | III. White box testing  |
| D. Alpha testing                  | IV. Performance testing |

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I

(B) A-III, B-I, C-IV, D-II

(C) A-IV, B-I, C-II, D-III

(D) A-III, B-IV, C-I, D-II

**Q72** Match List I with List II

| List I                    | List II                 |
|---------------------------|-------------------------|
| A. 8-Queens problems      | I. Divide and conquer   |
| B. Closest Pair of points | II. Dynamic Programming |
| C. String Edit problem    | III. Greedy Method      |
| D. Knapsack problem       | IV. Backtracking        |

Choose the correct answer from the options given below:

(A) A-I, B-II, C-III, D-IV

(B) A-I, B-IV, C-II, D-III

(C) A-IV, B-I, C-II, D-III

(D) A-II, B-III, C-IV, D-I

**Q73** Match List I with List II

| List I                                                                | List II      |
|-----------------------------------------------------------------------|--------------|
| A. Initial Configuration of stack and input in the LL Parsing process | I. \$ w\$    |
| B. Final Configuration of stack and input in LL Parsing process       | II. \$ \$    |
| C. Initial Configuration of stack and input in the LR Parsing process | III. \$S w\$ |
| D. Final Configuration of stack and input in the LL Parsing process   | IV. \$S \$   |

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-III, B-II, C-I, D-IV

(C) A-II, B-III, C-IV, D-I

(D) A-II, B-IV, C-I, D-III

**Q74** Match List I with List II

| List I             | List II                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------|
| A. Physical Layer  | I. Transforming the raw bits in the form of a frame for transmission                        |
| B. Data Link Layer | II. Control and monitoring of subnet                                                        |
| C. Network Layer   | III. Transmission of raw bits over a communication channel                                  |
| D. Transport Layer | IV. Datagrams transmit data through connection-oriented or connection-less using datagrams. |

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV

(B) A-II, B-III, C-I, D-IV

(C) A-III, B-I, C-II, D-IV

(D) A-II, B-IV, C-I, D-III

**Q75** Match List I with List II

| List I           | List II                  |
|------------------|--------------------------|
| A. Regression    | I. Unsupervised Learning |
| B. KNN           | II. Supervised Learning  |
| C. Clustering    |                          |
| D. Decision Tree |                          |

Choose the correct answer from the options given below:

(A) A-I, B-II, C-I, D-II

(B) A-II, B-II, C-I, D-II

(C) A-I, B-II, C-II, D-I

(D) A-II, B-II, C-II, D-I

**Q76** Let 'G' be a simple, undirected graph with  $n$  ( $n > 10$ ) vertices and  $k$  connected components. What is the minimum possible number of edges of G?

(A)  $k(n - 1)$ (B)  $n(k - 1)$ (C)  $n + k - 2$ (D)  $n - k$ 

**Q77** The sequence of events that happen during a typical fetch operation is/are:

- A. Memory
- B. MBR
- C. MAR
- D. PC
- E. IR

Choose the correct order from the options given below:

- (A) A, B, C, D, E
- (B) D, C, A, B, E
- (C) E, D, C, B, A
- (D) D, C, A, E, B

**Q78** Arrange the following search and sort algorithms in ascending order of their worst-case complexities.

- A. Merge sort
- B. Linear Search
- C. Bubble Sort
- D. Binary Search

Choose the correct answer from the options given below:

- (A) D, B, A, C
- (B) B, D, A, C
- (C) B, D, C, A
- (D) B, A, D, C

**Q79** Consider the following schedule S.

| T1   | T2   | T3   | T4   | Timestamp |
|------|------|------|------|-----------|
|      | R(X) | R(Y) |      | t1        |
|      | W(X) | W(Y) |      | t2        |
| R(Y) |      |      | R(X) | t3        |
| W(Y) |      |      | W(X) | t4        |
|      |      | R(X) |      | t5        |
|      |      | W(X) |      | t6        |

(R(X)) denotes the read operation on data item X by transaction Ti.

W(X) denotes write operation on data item X by transaction Ti.

Choose the correct execution order for the above schedule S.

- (A) T3-T2-T1-T4
- (B) T2-T3-T1-T4
- (C) T2-T4-T3-T1
- (D) T4-T2-T3-T1

**Q80** A disk has 300 cylinders ( 0 to 299 ). If the initial position of the read-write head is at cylinder 101, moving towards the higher end of the disk,

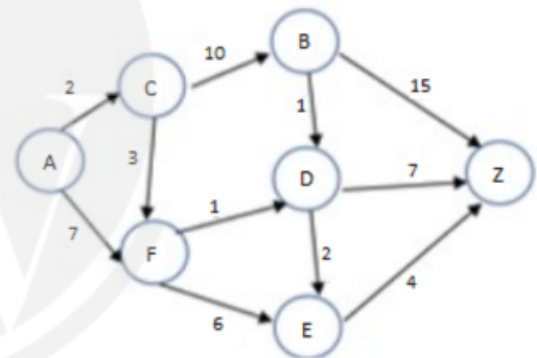
calculate the total head movement following C\_LOOK policy for the read request sequence: 123, 45, 197, 87, 250, 198, 280

- (A) 456
- (B) 450
- (C) 458
- (D) 445

**Q81** The incremental model of software development is \_\_\_\_

- (A) A reasonable approach when requirements are well defined.
- (B) A good approach when a working core product is required quickly.
- (C) The best approach to use for projects with large development teams.
- (D) A revolutionary model that is not used for commercial products.

**Q82** Consider the graph given below. What is the order in which the nodes mentioned in the options are to be traversed to find the shortest path from A to Z using best first search?



- A. Node-A
- B. Node-B
- C. Node-C
- D. Node-D
- E. Node-E

Choose the correct answer from the options given below:

- (A)  $A \rightarrow C \rightarrow F \rightarrow D \rightarrow E$
- (B)  $A \rightarrow C \rightarrow E \rightarrow B$
- (C)  $A \rightarrow C \rightarrow F \rightarrow E \rightarrow B$
- (D)  $A \rightarrow C \rightarrow D \rightarrow F$

**Q83** The precedence of regular expression operators from lowest to highest precedence is:

- (A) Star, Concatenation, Union



- (B) Union, Star, Concatenation  
 (C) Union, Concatenation, Star  
 (D) Star, Union, Concatenation

**Q84** Arrange the following in the correct order which respect to key generation for RSA algorithm.

- A. Choose an integer  $e$  such that  $1 < e < \lambda(n)$   
 B. Compute Key length  $n = p * q$   
 C. Find  $d$  which is equal to  $1/e$   
 D. Choose two large prime numbers  $p$  and  $q$   
 E. Compute  $\lambda(n)$

Choose the correct answer from the options given below:

- (A) A-B-C-D-E                      (B) D-C-B-E-A  
 (C) D-B-E-A-C                      (D) A-B-D-E-C

**Q85** Which of the following is related to function of layers in a neural network?

- A. Information  $\rightarrow$  Input layer  $\rightarrow$  Hidden layer  
 B. Bias  $\rightarrow$  output layer  $\rightarrow$  hidden layer.  
 C. Hidden layer  $\rightarrow$  algorithm  $\rightarrow$  output layer.  
 D. Input layer  $\rightarrow$  output layer  $\rightarrow$  algorithms.

Choose the correct answer from the options given below:

- (A) Only sequence 'A' is correct.  
 (B) Only sequence 'B' is correct.  
 (C) Only sequence 'C' is correct.  
 (D) Only sequence 'D' is correct.

**Q86** Given below are the two statements:

Statement I: The main objective for using virtual memory is to increase the effective capacity of the memory system

Statement II : The size of virtual memory can be larger than the main memory

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true.  
 (B) Both Statement I and Statement II are false.  
 (C) Statement I is true but Statement II is false.  
 (D) Statement I is false but Statement II is true.

**Q87** Given below are the two statements:

Statement I : Any retrieval request that is specified in the basic relational algebra can also be specified in relational calculus.

Statement II : Any retrieval request that is specified in the relational calculus can also be specified in basic relational algebra.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both Statement I and Statement II are true.  
 (B) Both Statement I and Statement II are false.  
 (C) Statement I is true but Statement II is false.  
 (D) Statement I is false but Statement II is true.

**Q88** The primary objectives of the SWE-IPT are to:

- A. Solicit stakeholder needs and expectations.  
 B. Specify the software requirements for the product and post-development processes.  
 C. Generate the integrated technical plans for accomplishing the software development effort.  
 D. Translate stakeholder requirements into a complete, specified architectural description.  
 E. Ensure that the software technical data package is sufficiently detailed to facilitate an efficient and effective software implementation.

Choose the correct answer from the options given below:

- (A) All the statements are correct.  
 (B) Only B & C are correct.  
 (C) Only A & E are correct.  
 (D) Only C, D & E are correct.

**Q89** Consider the following statements:

- (A) For every regular language, we can design Turing Machine  
 (B) For every context free language, we can design Turing Machine

Which of the following is correct?

- (A) Statement-A is false and Statement-B is true.  
 (B) Statement-A and Statement-B both are false.  
 (C) Statement-A is true and Statement- B is false.  
 (D) Statement-A and Statement-B both are true.

**Q90** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

A: Artificial Neural Networks are the backbone of deep learning algorithms.

R. Back propagation is one of the techniques used to train an ANN.



In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both A and R are correct and R is the correct explanation of A.
- (B) Both A and R are correct but R is NOT the correct explanation of A.
- (C) A is correct but R is not correct.
- (D) A is not correct but R is correct.

**Directions (91-95) Read the following passage and answer the given questions.**

Read the passage carefully and answer the question that follows

For a knapsack problem,  $P$  and  $W$  are profits and weights for a set of 7 items as given below:

$$P = \{9, 5, 2, 7, 6, 16, 9\}$$

$$W = \{2, 5, 6, 11, 1, 9, 3\}$$

**Q91** Find the optimal solution for fractional knapsack problem when its maximum capacity is 15 .

- (A) 43                      (B) 40
- (C) 45                      (D) 44

**Q92** What is the maximum value of items you can carry if maximum weight allowed is 25 with a fractional knapsack?

- (A) 48.1                      (B) 49
- (C) 47.9                      (D) 40

**Q93** If new item ( $P = 5, W = 2$ ) is added to the list, total items is 8 and capacity is 25 , then the best-case time of fractional knapsack for this case is \_\_\_\_

- (A)  $O(8)$                       (B)  $O(64)$
- (C)  $O(24)$                       (D)  $O(56)$

**Q94** The most efficient solution for the fractional knapsack problem with maximum capacity 15 can be obtained by using. \_\_\_\_

- (A) Divide and Conquer
- (B) Dynamic Programming
- (C) Greedy Algorithm

(D) Backtracking

**Q95** Which of the following statements is correct for a knapsack?

- (A) 0/1 knapsack can be applied for all kinds of problems but fractional knapsack can be applied only if the problem meets certain conditions.
- (B) There is no difference between the two. They give the same solution for any problem.
- (C) We use the greedy algorithm for 0 / 1 knapsack and dynamic programming for fractional knapsack.
- (D) In 0/1 knapsack problem, items cannot be divided into smaller portions and in fractional knapsack we can divide the item into required proportions of weight.

**Directions (96-100) Read the following passage and answer the given questions.**

Use the following schema of the academic institution relational database:

student (rollno, name, degree, year, sex, deptno, advisor)

department (deptid, name, hod, phone)

professor (empid, name, sex, startyear, deptno, phone)

course (courseid cname, credits, deptno)

enrollment (rollno, courseid, sem, year, grade)

teaching (empid, courseid, sem, year, classroom)

prerequisite (precourseid, courseid)

deptno is foreign key in the student, professor and course relations referring to deptid of department relation; advisor is a foreign key in the student relation referring to empid of professor relation; hod is a foreign key in the department relation referring to empid of professor relation; rollno is a foreign key in the enrollment relation referring to rollno of student relation; courseid is a foreign key in the enrollment, teaching relations referring to courseid of course relation; empid is a foreign key of the teaching relation referring to empid of professor.



professor relation; precourseid and courseid are foreign keys in the prerequisite relation referring to courseid of the course relation;

**Q96** Which of the following queries would retrieve the roll number and names of students who have enrolled for all the prerequisite courses of the course number 324 ?

- (A) Select s.rollno, s.name From student s, enrollment e Where e.rollno = s.rollno and e.courseid = ANY (select precourseid from prerequisite p where p.courseid = "324")
- (B) Select s.rollno, s.name From student s, enrollment e Where e.rollno =s.rollno and e.courseid = ALL (select precourseid from prerequisite p where p.courseid = "324")
- (C) Select s.rollno, s.name From student s Where NOT EXISTS (select \* prerequisite p where p.courseid = " 324 ") and NOT EXISTS (select \* from enrollment e where e.courseid = p.precourseid and e.rollno = s.rollno))
- (D) Select s.rollno, s.name From student s Where EXISTS (select \* prerequisite p where p.courseid = "324") and EXISTS (select \* from enrollment e where e.courseid = p.precourseid and e.rollno = s.rollno))

**Q97** Given query: select s.rollno from students where s.sex = ALL (select f.sex from professor f where f.empid = s.advisor).

Consider the following statements about the given query that retrieves the set of all students whose gender is same as the gender of their advisor:

- S1. It has a correlated subquery
- S2. It has an uncorrelated subquery
- S3. If we replace ALL by ANY in the above query, its result will be different
- S4. If we replace ALL by ANY in the above

query, its result will be same.

Identify all the statement that are TRUE:

- (A) S1 and S3 are true
- (B) S1 and S4 are true
- (C) S2 and S3 are true
- (D) S2 and S4 are true

**Q98** Which of the following queries would compute, for each department, the total credits of all the courses offered by the department?

- (A) select deptid, name, sum(credits) as totalcredits from department, course where deptid=deptno group by deptid, name;
- (B) select deptid, name, count(credits) as totalcredits from department, course where deptid=deptno
- (C) group by deptid, name; select deptid, name, sum(credits) as totalcredits from department, course
- (D) select deptid, name, sum(credits) as totalcredits from department, course group by deptid, name having deptid=deptno;

**Q99** Which of the following queries would retrieve the dept id and name of all the departments that are such that the total of the credits of all the offered courses by the department is strictly greater than 40 ?

- (A) select deptid, name, sum(credits) as totalcredits from department, course where totalcredits >40 group by deptid, name having deptid = deptno;
- (B) select deptid, name, sum(credits) as totalcredits from department, course where deptid = deptno and totalcredits >40 group by deptid, name;
- (C) select deptid, name, sum(credits) as totalcredits



from department, course  
 group by deptid, name  
 having deptid = deptno and  
 totalcredits >40

- (D) select deptid, name, sum(credits) as  
 totalcredits from department, course  
 where deptid = deptno  
 group by deptid, name  
 having totalcredits >40

**Q100 Given query:**

```
select s.rollno, s.name
from student s
where EXISTS (select c.courseid from
course c Where c.deptno =5 and
EXISTS (select e.* from enrollment e
where e.courseid = c.courseid and
e.rollno = s.rollno));
```

Which one of the following statements is correct with respect to the given query?

- (A) It retrieves the roll number and names of students who have not enrolled for all course offered by department number 5 .
- (B) It retrieves the roll number and names of students who have enrolled for all course offered by department number 5 .
- (C) It retrieves the roll number and names of students who have enrolled for at least one course offered by department number 5
- (D) It retrieves the roll number and names of students who are such that there exists at least one course offered by department number 5 which is not enrolled by the student.



## Answer Key

---

Q1 (D)  
Q2 (C)  
Q3 (B)  
Q4 (C)  
Q5 (C)  
Q6 (B)  
Q7 (A)  
Q8 (B)  
Q9 (C)  
Q10 (B)  
Q11 (D)  
Q12 (C)  
Q13 (B)  
Q14 (A)  
Q15 (B)  
Q16 (C)  
Q17 (C)  
Q18 (B)  
Q19 (A)  
Q20 (B)  
Q21 (D)  
Q22 (C)  
Q23 (B)  
Q24 (C)  
Q25 (D)  
Q26 (D)  
Q27 (B)  
Q28 (C)  
Q29 (B)  
Q30 (D)  
Q31 (B)

Q32 (D)  
Q33 (B)  
Q34 (C)  
Q35 (B)  
Q36 (B)  
Q37 (C)  
Q38 (C)  
Q39 (D)  
Q40 (A)  
Q41 (A)  
Q42 (B)  
Q43 (D)  
Q44 (B)  
Q45 (B)  
Q46 (C)  
Q47 (A)  
Q48 (B)  
Q49 (B)  
Q50 (B)  
Q51 (D)  
Q52 (A)  
Q53 (C)  
Q54 (A)  
Q55 (B)  
Q56 (C)  
Q57 (C)  
Q58 (A)  
Q59 (D)  
Q60 (A)  
Q61 (A)  
Q62 (D)



Q63 (B)  
Q64 (D)  
Q65 (A)  
Q66 (B)  
Q67 (B)  
Q68 (B)  
Q69 (B)  
Q70 (D)  
Q71 (B)  
Q72 (C)  
Q73 (B)  
Q74 (C)  
Q75 (B)  
Q76 (D)  
Q77 (B)  
Q78 (A)  
Q79 (B)  
Q80 (A)  
Q81 (B)

Q82 (A)  
Q83 (C)  
Q84 (C)  
Q85 (A)  
Q86 (A)  
Q87 (A)  
Q88 (A)  
Q89 (D)  
Q90 (B)  
Q91 (B)  
Q92 (A)  
Q93 (C)  
Q94 (C)  
Q95 (D)  
Q96 (B)  
Q97 (A)  
Q98 (A)  
Q99 (D)  
Q100 (C)



## Hints & Solutions

**Q1 Text Solution:**

Option- **D** ( $\neg Q \rightarrow P$ )

**Explanation:**

$P \rightarrow Q = \neg P \vee Q$  which is also equal to  $\neg(P \wedge \neg Q)$  and  $P \rightarrow \neg Q$

**A.**  $\neg P \vee Q = P \rightarrow Q$  which is logically equivalent.

**B.**  $\neg Q \rightarrow \neg P = Q \vee \neg P = \neg P \vee Q = P \rightarrow Q$  which is logically equivalent.

**C.**  $\neg(P \wedge \neg Q) = \neg P \vee Q = P \rightarrow Q$  which is logically equivalent.

**D.**  $\neg Q \rightarrow P = Q \vee P$  which is not logically equivalent.

**Q2 Text Solution:**

Option - **C (1806)**

**Explanation:**

This problem can be addressed with Stirling numbers of the second kind and the multinomial coefficient.

$$S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$$

First, divide the 7 people into three non-empty groups (using Stirling numbers of the second type):

$$S(7, 3) = 3 \cdot S(6, 3) + S(6, 2)$$

Similarly, calculating and assigning values of  $S(6, 3)$  and  $S(6, 2)$  we got,  $S(7, 3) = 301$

Each of these three groups should be assigned a floor. This can be done in  $3! = 6$  ways.

The total number of ways,  $301 \times 6 = 1806$

**Q3 Text Solution:**

Option: **B(210)**

**Explanation:** Given that, we have to find the coefficients of  $x^2 y^3 z^2$  by expanding  $(x+y+z)^7$ .

The general expansion of  $(x+y+z)^7$  can be written as  $(x+y+z)^7 = \sum \frac{7!}{n_1! n_2! n_3!} x^{n_1} y^{n_2} z^{n_3}$

where,  $n_1=2, n_2=3, n_3=2$

and,  $n_1+n_2+n_3=n$  which is,  $2+3+2=7$

Thus, co-efficient =  $7! / (2! 3! 2!) = 5040 / 24 = 210$

**Q4 Text Solution:**

Option- **C(36)**

**Explanation:** To ensure that at least six students get the same mark, we use the pigeonhole principle.

There are seven potential grades. If each grade is allotted to at most 5 pupils, there may be:  $7 \times 5 = 35$  students

However, adding one additional student (totaling 36) requires one of the grades to be given to at least six pupils.

Thus, the minimal number of pupils needed is: 36

**Q5 Text Solution:**

Option- **C (1111 1111 1110 0100)**

**Explanation:** In order to turn the decimal number 28 into a 16-bit 2's complement representation, we do the following steps:

**Step 1: Convert 28 into binary:**  $28 = (11100)_2$

We now need to express 28 in a 16-bit format by padding it with leading zeros:  $28 = (0000\ 0000\ 0001\ 1100)_2$

**Step 2: Calculate 2's Complement:** 2's

complement of 00000000000111100 would be:

**1111 1111 1110 0100**

**Q6 Text Solution:**

Option: **B ( $A'B + CD$ )**

**Explanation:** Given,  $F(A, B, C, D) = \sum (3, 4, 5, 6, 7, 11, 15)$  which is 4 variable K-Map.

|      | C'D'  | C'D   | CD     | CD'   |
|------|-------|-------|--------|-------|
| A'B' |       |       | (3) 1  |       |
| A'B  | (4) 1 | (5) 1 | (7) 1  | (6) 1 |
| AB   |       |       | (15) 1 |       |
| AB'  |       |       | (11) 1 |       |

Grouping them we found two groups with 4 entries.

Group 1 consists of 3, 7, 11, and 15 and Group 2



consists of 4,5,6, and 7

After simplification,  $F(A,B,C,D) = A'B + CD$

**Q7 Text Solution:**

Option - A (011,101,000)

**Explanation:** The input sequence for the specified flipflop-based state counter is as follows:

$$T_P = R$$

$$T_Q = P'$$

$$T_R = Q'$$

$$P^+ = R \oplus P$$

$$Q^+ = P' \oplus Q$$

$$R^+ = Q' \oplus R$$

| Initial State |   |   | Current Input |       |       | Next State |       |       |
|---------------|---|---|---------------|-------|-------|------------|-------|-------|
| P             | Q | R | $T_P$         | $T_Q$ | $T_R$ | $P^+$      | $Q^+$ | $R^+$ |
| 0             | 0 | 0 | 0             | 1     | 1     | 0          | 1     | 1     |
| 0             | 1 | 1 | 1             | 1     | 0     | 1          | 0     | 1     |
| 1             | 0 | 1 | 1             | 0     | 1     | 0          | 0     | 0     |

In a flip-flop, low input (0) results in the next state being the current state (Q), whereas high input (1) toggles/complements the existing state( $Q'$ )

With PQR as 000, the next state would be current input, i.e, 011

With PQR as 011, the next state would depend on the input values and will give 101 as the next state

With PQR as 101, the next state again would depend on input values and will give 000 as the next state.

Thus, the next three states would be: 011-101-000

**Q8 Text Solution:**

Option- B(  $R1 = 1100$ ,  $R2 = 1010$  )

**Explanation:** The numbers in 2's Complement form range from -8 to +7.

**Option B.  $R1 = 1100$  and  $R2 = 1010$**

$R1 + R2 = (-4) + (-6) = -10$  [ Not within range, hence Overflow ]

In options A, C, and D, the solution is within the range and has no overflow.

**Q9 Text Solution:**

Option- C(Internal functions are known at link time)

**Explanation:** This statement is false because internal functions (functions written inside the same source code or module) are resolved at compile time rather than link time.

**Q10 Text Solution:**

Option - B (  $*(*(arr + 2) + 3) + 1$  )

**Explanation:** Given Array,

```
int arr[3][4][2] = {
 {{2, 4}, {7, 8}, {3, 4}, {5, 6}},
 {{7, 6}, {3, 4}, {5, 3}, {2, 3}},
 {{8, 9}, {7, 2}, {3, 4}, {5, 1}}
};
```

The array can be seen as:

$arr[0] = \{ \{2, 4\}, \{7, 8\}, \{3, 4\}, \{5, 6\} \}$

$arr[1] = \{ \{7, 6\}, \{3, 4\}, \{5, 3\}, \{2, 3\} \}$

$arr[2] = \{ \{8, 9\}, \{7, 2\}, \{3, 4\}, \{5, 1\} \}$

In a 3D array:

$arr[0]$  represents the initial block (the first 4x2 matrix).

$arr[0][0]$  denotes the first row of the first block: {2, 4}.

$arr[0][0][1]$  represents the second element in this row, which is 4.

**Option B. Simplifying:**

$arr + 2$ : This advances the pointer to the third block ( $arr[2]$ ).

$*(arr + 2)$ : Dereferencing this yields the array  $arr[2]$ , which contains  $\{ \{8, 9\}, \{7, 2\}, \{3, 4\}, \{5, 1\} \}$ .

$*(*(arr + 2) + 3)$ : This accesses the fourth row of  $arr[2]$ : {5, 1}.

$*(*(arr + 2) + 3) + 1$ : This accesses the second element in the fourth row, which is one.

**Q11 Text Solution:**

Option - D (Run time polymorphism)

**Explanation:** Runtime polymorphism occurs when various derived classes are utilized interchangeably via an interface given by a shared base class. Runtime polymorphism determines the method to be invoked depending on the actual type of the object rather than the type of the reference or pointer.



**Q12 Text Solution:**

Option- **C( Bezier Curve)**

**Explanation:** In a Bezier curve, the polygon's end and beginning vertices are on the curve, but the remaining vertices (also known as control points) define the curve's derivatives, order, and shape.

The curve is defined by a set of control points, and its degree is equal to the number of control points that are less than one.

**Q13 Text Solution:**

Option- **B(Union)**

**Explanation:** The Union operator joins the rows (tuples) of two relations (tables), resulting in a new relation that contains all rows from both input relations, with duplicates eliminated. For the operation to be valid, the two relations must have the same amount of characteristics (columns), as well as comparable data types.

**Q14 Text Solution:**

Option- **A(P,Q,R,S)**

**Explanation:** Given, relation R (P, Q, R, S, T, U, V, W, X,Y) and FD's:

$\{P, Q, S\} \rightarrow T$

$R \rightarrow Y$

$\{P,Q\} \rightarrow V$

$\{R, X\} \rightarrow X$

$Q \rightarrow U$

$V \rightarrow \{W, X\}$

By taking attribute closure of the options given below, we found:

Candidate key:  $\{P, Q, R, S\}^+ = \{P, Q, R, S, T, U, V, W, X, Y\}$

**Q15 Text Solution:**

Option- **B(Level 1)**

**Explanation:** Given the need for large database updates (which imply write-intensive activities) and the necessity for fault tolerance in log file storage.

RAID 1 offers fault tolerance by replicating data on two drives. This is especially crucial for log files since data integrity must be maintained even if a disk fails.

**Q16 Text Solution:**

Option- **C(The internal nodes in a branch are connected by AND and branches by OR)**

**Explanation:** When transforming a decision tree to logical rules, internal nodes are linked using AND since each condition along the route must be true for the rule to apply. The branches are linked together by OR because they represent different criteria that result in the same class label. In essence, AND assures that every requirement on a path must hold, whereas OR combines many pathways that lead to the same result.

**Q17 Text Solution:**

Option- **C ( Access Matrix )**

**Explanation:** The access matrix is the most common paradigm for managing and regulating resource access in computer security and protection frameworks. This matrix aids in the modeling of access control rules, in which each subject is assigned a specific set of rights for various objects.

**Q18 Text Solution:**

Option- **B(Dynamically allocate the memory during process run time)**

**Explanation:** The heap region of a process in memory is responsible for dynamic memory allocation. It is where memory is allocated at runtime by methods such as malloc() in C and new in C++. The heap enables a program to allocate memory for variables whose sizes are unknown at compile time and might change during execution.

**Q19 Text Solution:**

Option- **A(8)**

**Explanation:** Given, reference string 1,2,3,4,5,3,4,1,6,7,8,7,8 with 4 frames

The **Optimal Page Replacement Algorithm** replaces the page that will not be used for the longest period of time in the future.Using the Optimal Page Replacement policy, the below table shows page hit/miss  
Step-by-step analysis:



| Step | Reference | Frames       | Hit/Miss | Explanation                                         |
|------|-----------|--------------|----------|-----------------------------------------------------|
| 1    | 1         | [1]          | Miss     | The first reference, page 1 is loaded.              |
| 2    | 2         | [1, 2]       | Miss     | The first reference, page 2 is loaded.              |
| 3    | 3         | [1, 2, 3]    | Miss     | The first reference, page 3 is loaded.              |
| 4    | 4         | [1, 2, 3, 4] | Miss     | The first reference, page 4 is loaded.              |
| 5    | 2         | [1, 5, 3, 4] | Miss     | Page 2 is replaced by page 5 (optimal replacement). |
| 6    | 3         | [1, 5, 3, 4] | Hit      | Page 3 is already in memory.                        |
| 7    | 4         | [1, 5, 3, 4] | Hit      | Page 4 is already in memory.                        |
| 8    | 1         | [1, 5, 3, 4] | Hit      | Page 1 is already in memory.                        |
| 9    | 1         | [6, 5, 3, 4] | Miss     | Page 1 is replaced by page 6 (optimal replacement). |
| 10   | 5         | [6, 7, 3, 4] | Miss     | Page 5 is replaced by page 7 (optimal replacement). |
| 11   | 3         | [6, 7, 8, 4] | Miss     | Page 3 is replaced by page 8 (optimal replacement). |
| 12   | 7         | [6, 7, 8, 4] | Hit      | Page 7 is already in memory.                        |
| 13   | 8         | [6, 7, 8, 4] | Hit      | Page 8 is already in memory.                        |

Total Page faults=8

#### Q20 Text Solution:

Option- **B(one-to-one > many-to-many > many-to-one)**

**Explanation:** In terms of concurrency,

-One-to-one provides the highest degree of concurrency and is the most efficient in systems with multiple processors (or cores).

-Many-to-many balance between concurrency and resource management, moderate concurrency.

-Many-to-One offers low concurrency and is not ideal for systems that require high responsiveness or true parallelism

#### Q21 Text Solution:

Option - **D(Write self-documenting code, not program documentation)**

**Explanation:** While self-documenting code (clean, readable code with understandable variable and function names) is a desirable practice, the notion of not using program documentation does not always apply. Program documentation (comments, documentation files, and user manuals) is still useful in many situations, particularly for complicated code or systems. It helps other developers (and even your future self) comprehend the code's context, logic, and purpose.

Documentation is critical for expressing the "why" behind the code, which self-documenting code may not adequately convey.

#### Q22 Text Solution:

Option-**C( $RE = P * L$ )**

**Explanation:** Risk exposure (RE) is a risk management concept that estimates the possible loss if a risk event occurs. It is computed by multiplying the risk's probability (P) by the potential loss (L) if the event occurs.

P represents the likelihood of the risk occurring (a number between 0 and 1).

L represents the loss (or effect) that would occur if the risk occurred.

The formula for risk exposure (RE) is:  $RE = P * L$

#### Q23 Text Solution:

Option- **B(Clear-Box wrapping)**

**Explanation:** Clear-box wrapping aims to construct a wrapper over a component that allows insight into its underlying structure and logic while keeping the component's exterior behavior unchanged.

Clear-box wrapping exposes the component's underlying structure, which the wrapper may interact with or alter in order to give more functionality or adapt the component to a new system.

#### Q24 Text Solution:



Option- C (Independent logic paths in the program)

**Explanation:** Cyclomatic complexity is the number of independent pathways through the program's control flow. The measure helps determine how many distinct pathways exist, which is critical for maintaining proper test coverage.

**Q25 Text Solution:**

Option- D(Unbalanced Binary Tree)

**Explanation:** If the difference exceeds 1 (i.e., the height difference between the left and right subtrees is greater than 1), the tree is said to be **unbalanced**.

In an unbalanced binary search tree, the left and right subtrees for any given node differ in height by more than one

**Q26 Text Solution:**

Option- D(Compression, frequency)

**Explanation:** Fast Fourier Transforms (FFT) are approaches for analyzing signals in the frequency domain rather than the time domain. They work by breaking down complicated signals (such as digital audio or video) into their constituent frequencies. Furthermore, FFTs are extensively used in compression in digital signal processing. By translating a signal to the frequency domain, superfluous or redundant frequency components can be eliminated, resulting in a smaller data size.

**Q27 Text Solution:**

Option B is Correct

To solve this problem, a 2-tree (Huffman Tree) is constructed using the given weights, and the minimum weighted path length is calculated. In a Huffman Tree, it is necessary to repeatedly combine the two smallest weights until only one node remains.

Here, the weights are 4, 15, 25, 5, 8, 16

Steps to Build the Huffman Tree

1. Sort the weights in ascending order: 4, 5, 8, 15, 16, 25

2. Combine the two smallest weights repeatedly and keep track of the sum at each step. Each combined weight represents an internal node in the tree, and the total path length is the sum of these weights at different depths.

Step-by-Step Construction

- Combine 4 and 5  $\rightarrow 4 + 5 = 9$

New list: 8, 9, 15, 16, 25

- Combine 8 and 9  $\rightarrow 8 + 9 = 17$

New list: 15, 16, 17, 25

- Combine 15 and 16  $\rightarrow 15 + 16 = 31$

New list: 17, 25, 31

- Combine 17 and 25  $\rightarrow 17 + 25 = 42$

New list: 31, 42

- Combine 31 and 42  $\rightarrow 31 + 42 = 73$

New list: 73

Now, we sum all intermediate weights used in combining to calculate the weighted path length:

$$9 + 17 + 31 + 42 + 73 = 172$$

The minimum weighted path length is 172.

**Q28 Text Solution:**

Option- C(less than 1/2)

By the definition of universality, the probability that 2 given keys in the table collide under h is

$$1/m = 1/n^2. \text{ Since there are } \binom{n}{2} \text{ pairs of keys}$$

that can possibly collide, the expected number of collisions is

$$\binom{n}{2} \cdot \frac{1}{n^2} = \frac{n(n-1)}{2} \cdot \frac{1}{n^2} < \frac{1}{2}$$

**Q29 Text Solution:**

Option- B(1)

**Explanation:** The Pumping Lemma for Context-Free Languages says that each context-free language L has a constant p (dubbed the "pumping length") that allows any string s in L with length at least p to be divided into five parts:  $s = uvwxy$  where:

- $|vwx| \leq p$  (The substring vwx has a maximum length of p).



- For all  $i \geq 0$ , the string  $uv^iwx^iy$  is in the language  $L$ .
  - $|vx| \geq 1$  (i.e.,  $v$  and  $x$  must not be empty).
- Focus on  $vx$ : -  $vx$  is a crucial component of the string that is repeated (or pumped).
- The pumping lemma requires the minimum length of  $vx$  to be at least 1, as both  $v$  and  $x$  cannot be empty (because  $|vx| \geq 1$ ).

**Q30 Text Solution:**

Option- **Type-3**

Regular Grammars (Type-3): Regular grammars can generate the empty string ( $\epsilon$ ) because they can have rules such as  $A \rightarrow \epsilon$ , where a non-terminal produces the empty string. However, regular grammars are often  $\epsilon$ -free by design, particularly if they do not include production rules that directly derive  $\epsilon$  (the empty string). This is because regular grammars usually focus on generating finite strings and often exclude rules that would generate  $\epsilon$  unless specifically needed.

**Q31 Text Solution:**

Option- **B(From left to right)**

**Explanation:** The LR parser scans the input string from left to right.

It adds symbols to the stack and decreases when it discovers a matching production rule.

The procedure is repeated until the full string is analyzed and the parser reaches the start symbol.

**Q32 Text Solution:**

Option- **D (Minimum: 1,0,0,1; Maximum: 1, (n-1), (n-1), n)**

**Explanation:**

1. Initial state:

- A minimum of one beginning state.
- Maximum: one starting state.

2. Dead states:

- Minimum: zero.
- Maximum is  $n-1$ .

3. Unreachable states:

- Minimum: zero.
- Maximum is  $n-1$ .

4. Final (accepting) States:

- Minimum: at least one final state.
- Maximum:  $n$ .

**Q33 Text Solution:**

Option- **B(Address Resolution Protocol)**

**Explanation:** ARP is the mechanism for obtaining a host's MAC address from its IP address. It runs on the OSI model's Data Link Layer (Layer 2) and is used to translate IP addresses (Layer 3) into MAC addresses (Layer 2).

**Q34 Text Solution:**

**Data word:** 111111

**Remainder:** 110

**Codeword** (Data word + Remainder): 111111110

Option C is correct.

**Q35 Text Solution:**

In the Go-Back-N ARQ protocol, the sender maintains a window for unacknowledged packets. The window is defined by two pointers:  $S_f$  (Send Window First Outstanding Packet): This is the sequence number of the first unacknowledged packet.

$S_n$  (Send Window, the Next Packet to be Sent): This is the sequence number of the next packet to be sent.

Given:

$S_f = 3$

$S_n = 7$

The receiver sends an acknowledgment for  $AckNo = 6$ .

The  $AckNo = 6$  indicates that all packets up to packet 6 have been successfully received and acknowledged. Since the Go-Back-N protocol allows for sending packets up to  $S_n - 1$ , and the acknowledged packet is 6, the sender can now move the window forward.

The new  $S_f$  (the first unacknowledged packet) will move to  $AckNo + 1 = 6 + 1 = 7$ . The  $S_n$  (next packet to send) will also move forward to  $S_n = 7$  since  $S_n$  represents the next packet to be sent after the acknowledged ones.

Thus, after receiving the  $AckNo = 6$ , the new value of  $S_n$  is:  $S_n = 7$

The correct option is B.



**Q36 Text Solution:**

The Hamming distance between two code words (pieces of data) is simply the number of bit positions in which they differ. For example, if two binary numbers have a Hamming distance of 3, it means three bits are different between them.

**Error detection:**

1. If we want to detect **up to S errors** (meaning there could be S incorrect bits in the data), we need the minimum Hamming distance between any two valid code words to be **S + 1**.
2. If two valid code words differ by **S + 1 bits**, then even if **S bits** change (due to errors), the received word will still not match another valid code word. This allows detect that an error has occurred.

The correct option is B.

**Q37 Text Solution:**

A search algorithm is **complete** if it guarantees finding a solution if one exists.

A search algorithm is **optimal** if it finds the least-cost solution.

A heuristic function  $h(n)$  is **consistent** if, for every node  $n$  and every successor  $n'$  of  $n$  generated by any action  $a$ , the following inequality holds:  $h(n) \leq c(n, n') + h(n')$  where  $c(n, n')$  is the cost of reaching  $n'$  from  $n$ . This property is known as the **monotonicity** property.

**A\* Search:** Search is known to be **both complete and optimal** when the heuristic  $h(n)$  is consistent. Option C is correct.

**Q38 Text Solution:**

**C) Generalized Modus Ponens** is an inference rule used in logic and artificial intelligence that extends the basic modus ponens rule. Modus ponens is a simple rule of inference that states: If "P implies Q" ( $P \rightarrow Q$ ) is true and P is true, then Q must be true.

**Generalized Modus Ponens** goes by allowing the inference to apply to rules with variables. It allows us to capture more complex inference processes in a single rule, also when dealing with

logical statements involving variables and unification in knowledge-based systems.

**Q39 Text Solution:**

D) Many valued logic

Correct: Fuzzy logic is a form of many-valued logic that allows for values between true and false, rather than the binary choices of true (1) or false (0). Fuzzy logic can handle partial truth, where values can range continuously between completely true and completely false.

For example, in fuzzy logic, a statement like "The weather is warm" could have a truth value of 0.7, representing that it is somewhat true, rather than strictly true or false. This flexibility is particularly useful in systems that deal with uncertainty, like control systems and decision-making processes in artificial intelligence.

**Q40 Text Solution:**

The **4 Ws problem canvas** is a tool used in **problem scoping** to help identify the essential elements of a problem by addressing four key questions:

1. **Who** is affected by the problem?
2. **What** is the problem?
3. **Where** does the problem occur?
4. **Why** is it important to solve the problem?

This canvas helps in understanding the scope of the problem, setting clear boundaries, and identifying stakeholders, goals, and the context in which the problem exists. It is a crucial step in the initial stages of problem-solving, as it helps in defining and understanding the problem before moving on to data collection, modeling, and evaluation.

The correct option is A.

**Q41 Text Solution:**

**Statement B:** "A group cannot have any element which is idempotent except the identity element."

Correct. In a group, if an element  $g$  is idempotent (i.e.  $g * g = g$ ), then  $g$  must be the identity element. If  $g$  were not the identity, we could multiply both



sides by  $g^{-1}$ , yielding  $g=e$ , which is a contradiction unless  $g=e$  initially.

**Statement C:** "An isomorphism from a group  $\langle G, * \rangle$  to  $\langle G, * \rangle$  is called an automorphism."

Correct. An automorphism is defined as an isomorphism from a group to itself.

**Statement D:** "If every element in a group is its own inverse, then the group must be abelian."

Correct. If every element is its own inverse (i.e.  $g*g=e$  for all  $g$  in the group), then  $g*h=h*g$  for any two elements  $g$  and  $h$ . This is because:  $g*h=(g*h)^{-1}=h^{-1}*g^{-1}=h*g$  showing that the group is abelian.

Option A is correct.

#### Q42 Text Solution:

##### Statement A

Correct: A labeled binary tree is one where each node is distinct or uniquely labeled. For  $n$  labeled nodes, the number of possible labeled binary trees is given by:  $(2n)!/(n+1)!$ . This is derived from the Catalan number, which gives the number of distinct binary tree structures with  $n$  nodes, multiplied by  $n!$  (to account for labeling each node uniquely).

##### Statement B

Correct: In a binary search tree (BST), the structure of the tree is determined by the arrangement of values in such a way that each node's left subtree contains only nodes with values less than the node's value, and the right subtree contains only nodes with values greater.

- The number of unique BSTs with  $n$  distinct nodes is given by the  $n$ -th Catalan number, which is:  $C_n = (2n)!/(n+1)!n!$
- This formula counts the distinct ways to arrange  $n$  nodes in a BST

##### Statement D

Correct: In a simple, undirected, labeled graph with  $n$  vertices, each possible edge between a pair of vertices can either exist or not exist, independently of other edges.

- The number of edges in a complete graph with  $n$  vertices is  $n(n-1)/2$ .

- Since each of these edges can either be present or absent, there are  $2^{n(n-1)/2}$  possible configurations of edges, which represent all possible simple graphs for  $n$  labeled vertices.

Option B is correct.

#### Q43 Text Solution:

##### Reflexive:

A relation is reflexive if every element in the set has a pair  $(a,a)$  in the relation.

Here, we have  $(1,1)$ ,  $(3,3)$  and  $(4,4)$ , but we do not have  $(2,2)$ .

Since  $(2,2)$  is missing, the relation is **not reflexive**.

##### Symmetric:

A relation is symmetric if for any two elements  $(a,b) \in R$  then  $(b,a) \in R$

Here,  $(1,2) \in R$ , but  $(2,1) \notin R$ , so it's not symmetric.

##### Transitive:

A relation is transitive if for all  $(a,b)$  and  $(b,c)$  in  $R$ , the pair  $(a,c)$  is also in  $R$ .

We have  $(3,1)$  and  $(1,2)$  but there is no  $(3,2)$  that follows from this pair in terms of transitivity.

The relation fails to maintain transitivity for all cases, so it is **not transitive**.

Option D is correct answer.

#### Q44 Text Solution:

Taking Compliment of  $F$ , Hence

$$F' = ((X+Y+Z) (X'+Y) (Y'+Z))'$$

Using De Morgan's Law:

$$F' = (X+Y+Z)' + (X'+Y)' + (Y'+Z)'$$

Applying De Morgan's Law to each term Separately,

$$F' = X'Y'Z' + XY' + YZ' \text{ or } XY' + YZ' + X'Y'Z' \text{ (which is Option D)}$$

By taking  $Y'$  Common, we get,

$$F' = Y'[(X'Z' + X)] + YZ' \text{ (Since, Applying Distributive Law, } A+BC = (A+B)(A+C))$$

$$F' = Y'[(X+X')(X+Z')] + YZ'$$

$$F' = Y'[(X'Z') + X] + YZ' \text{ (Since } X+X' = 1)$$

$$F' = Y'[X+Z'] + YZ'$$

$$F' = XY' + Y'Z' + YZ'$$

Taking  $Z'$  Common,

$$F' = XY' + Z'(Y+Y')$$

$$F' = XY' + Z' \text{ (Since, } Y+Y' = 1) \text{ (which is Option B)}$$

Applying Distributive Law,



$F' = (X+Z')(Y'+Z')$  (which is Option C)

So, B, C, and D are equivalent to  $F'$ .

As for Option A,

Suppose,  $X=0$ ,  $Y=1$ ,  $Z=0$ ,

Now,  $F$  is equal to 0 as  $(Y'+Z)$  term will be zero.

Hence,  $F'$  must be equal to 1.

But Option A gives 0 as the term  $X+Y'$  evaluates to 0. So, Option A is not equal to  $F'$ .

#### Q45 Text Solution:

**Statement A: "The processor has direct access to both primary and secondary memory."**

**False.** The processor has direct access to **primary memory** (RAM), where active instructions and data are stored for quick access during execution. However, it does **not have direct access** to **secondary memory** (such as hard drives or SSDs).

**Statement B: "Primary memory stores the active instructions and data for the program being executed on the processor."**

**True.** Primary memory (RAM) holds the active data and instructions that the processor needs to execute programs. This is the main purpose of primary memory, providing fast access to the processor.

**Statement C: "Secondary memory is used as a backup memory."**

**True.** Secondary memory, such as hard drives, SSDs, or external storage, is used for long-term storage and acts as a backup to primary memory. It retains data even when the system is powered off.

**Statement D: "Memory system is implemented on a single level memory."**

**False.** Modern computer systems use a **multi-level memory system**, which includes primary memory (RAM), secondary memory (hard drives, SSDs), and often cache memory. This multi-level approach helps optimize speed and storage capacity.

Option B is correct answer.

#### Q46 Text Solution:

Inheritance in object-oriented programming provides the following benefits:

#### Run-time Polymorphism (B):

**True.** Inheritance enables run-time polymorphism, which allows derived classes to override methods of the base class. This is achieved through dynamic method, allowing different implementations to be called based on the object's runtime type.

#### Less Redundant Information (C):

**True.** Inheritance helps reduce redundancy by allowing derived classes to reuse the attributes and behaviors (methods) of a base class. This means that shared code does not need to be duplicated across multiple classes.

#### Dynamic Binding (D):

**True.** Inheritance supports dynamic binding, where method calls are resolved at runtime rather than compile-time. This enables polymorphism and allows derived classes to provide specific implementations of methods defined in the base class.

Correct option is C.

#### Q47 Text Solution:

**Option A: `string s1;`**

Correct: This is a declaration of an empty string `s1` using the default constructor.

**Option B: `string s2(s1);`**

Correct: This is a copy constructor, where `s2` is being initialized as a copy of `s1`.

**Option C: `string s3{7};`**

Incorrect: This syntax does not match any standard string constructor.

**Option D: `string s4{'a'};`**

Correct: This syntax initializes `s4` as a string containing a single character `'a'` using list initialization.

**Option E: `string s5{7, 'a'};`**

Correct: This uses a constructor to create a string with 7 occurrences of the character `'a'`.

The correct option is A.

#### Q48 Text Solution:

**Statement B: "The style sheets can stack on each other."**

**True.** In CSS (Cascading Style Sheets), styles from different sheets can be stacked. This allows



multiple stylesheets to be applied to the same document, with each stylesheet contributing to the overall appearance. The HR department's stylesheet can be stacked on top of the corporate stylesheet to add specific differences.

**Statement C: "The style sheets can override each other."**

**True.** CSS allows stylesheets to override each other based on the order in which they are loaded. If the HR department's stylesheet contains rules that differ from the corporate stylesheet, these rules can override the corporate styles for elements specified in both stylesheets.

Option B is correct answer.

**Q49 Text Solution:**

The transaction  $T_1$  involves transferring Rs. 50 from account A to account B, with the following operations:

Step 1: Read(A) – Reads the balance of account A.

Step 2:  $A := A - 50$  – Subtracts 50 from the balance of account A in memory.

Step 3: Write(A) – Write the updated balance of account A back to the database.

Step 4: Read(B) – Reads the balance of account B.

Step 5:  $B := B + 50$  – Adds 50 to the balance of account B in memory.

Step 6: Write(B) – Write the updated balance of account B back to the database.

After the 3rd step (which is Write(A)) but before the 5th step (which is  $B := B + 50$ ), the database has only partially completed the transaction:

The balance in account A has been deducted by 50 and written back to the database.

However, the corresponding addition to account B has not yet occurred.

Database Status:

The database is temporarily inconsistent because Rs. 50 has been deducted from account A but not yet added to account B. This partial update means the total amount between the two accounts does not reflect the intended transfer.

The correct option is B.

**Q50 Text Solution:**

Canonical cover of a set of functional dependencies, for this make sure to remove any redundant dependencies. Redundancy occurs when a functional dependency can be derived from other dependencies in the set.

FD1 is not redundant as  $(\text{StudID})^+$  without FD1 does not contain Dept. FD2 is redundant as  $(\text{Dept})^+$  remains the same without FD2. Hence, it can be removed. Now we have FD1, FD3, FD4, FD5.

FD3 is redundant as  $(\text{StudID})^+$  remains the same without FD3. Hence, it can be removed. Now we have FD1, FD4, FD5. The set cannot be further reduced. So, the final reduced set is FD1, FD4, FD5. If the above procedure starts by rearranging the set and placing FD5 at the beginning, the canonical cover is reduced to FD1, FD2, FD3, FD4. Hence, both options (A) and (C) are correct.

**Q51 Text Solution:**

**Statement A: "If the time quantum is extremely small, round robin can result in a large number of context switches."**

**True.** In Round Robin scheduling, each process is given a small amount of CPU time (time quantum) before switching to the next process. If the time quantum is too small, the CPU will frequently switch between processes, resulting in a large number of context switches. This can lead to overhead and inefficiency.

**Statement B: "The average turnaround time of a set of processes does not necessarily improve as the time-quantum size increases."**

**True.** Increasing the time quantum does not always lead to a lower average turnaround time. If the time quantum is too large, Round Robin effectively behaves like a First-Come, First-Served (FCFS) scheduling algorithm, which might not provide a good average turnaround time.

**Statement C: "The average turnaround time can be improved if most processes finish their next CPU burst in a single time quantum."**



**True.** If the time quantum is chosen such that it allows most processes to complete their CPU bursts within a single quantum, the system can reduce waiting times and context switches, improving the average turnaround time.

Option D is correct.

**Q52 Text Solution:**

Option A is Correct

The scheduling Algorithm being used is Shortest Remaining Time First

The GANTT Chart Formed will be

| A   | B   | C   | D    | E     | F     |
|-----|-----|-----|------|-------|-------|
| 0-3 | 3-5 | 5-8 | 8-12 | 12-15 | 15-21 |

Since, Turnaround Time = Completion Time – Arrival Time

| Process | Arrival Time | Execution Time | Completion Time | Turnaround Time |
|---------|--------------|----------------|-----------------|-----------------|
| A       | 0            | 6              | 8               | 8               |
| B       | 3            | 2              | 5               | 2               |
| C       | 5            | 4              | 12              | 7               |
| D       | 7            | 6              | 21              | 14              |
| E       | 10           | 3              | 15              | 5               |

Average Turnaround Time =  $\frac{(8+2+7+14+5)}{5} = 7.2$  ms

**Q53 Text Solution:**

The basic page replacement process follows these steps in order:

C. Location of desired frame is found from disk:  
This step happens first for finding the location of the page to be replaced. In some systems, the operating system may first locate the desired page (or the page to be swapped in) on disk before taking any other action.

A. Page replacement algorithm is used in case there is no free frame:

After finding the desired page on disk, the page replacement algorithm is invoked to decide which page in memory will be replaced if there are no free frames available. This happens when there's a page fault.

D. Desired page is brought into newly freed frame and page & frame tables are updated:  
Once the replacement page is identified, the desired page is loaded into the newly freed frame, and the relevant page and frame tables are updated to reflect this change.

B. The process is continued by restarting the instruction caused by TRAP:

Finally, after the page is loaded, the process continues by restarting the instruction that caused the page fault, ensuring that the program can continue execution from where it was interrupted.

**Q54 Text Solution:**

**A) Both Statement I and Statement II are true.**

**1. Statement I: "Software testing must be recognized as a significant effort, and its planning must begin at the earliest possible time."**

- **True.** Software testing is essential for ensuring the quality and reliability of the final product. Planning for testing early in the software development lifecycle helps identify potential issues sooner, reduces the probability of defects later on, and allows for a smoother and more efficient testing process.

**2. Statement II: "The cost of software testing is an important contribution in determining overall software development costs."**

- **True.** Software testing costs are a significant part of the total development cost. Testing requires resources, time, and effort to ensure that the software meets requirements and is free of defects. These costs directly impact the overall budget, making them an essential factor in calculating the full cost of software development.

**Q55 Text Solution:**

In **unit testing**, the focus is generally on testing individual units of code to verify that they work



as expected under various conditions.

### Algorithmic Performance

**Need not be assessed** in unit testing. While performance may be a concern in the broader testing process, it is not part of unit testing. Unit testing primarily focuses on correctness rather than performance.

### Number of Lines of Code

**Need not be assessed** in unit testing. The physical length of code (i.e., the number of lines) is irrelevant to the functionality being tested and does not provide insight into whether the unit functions correctly.

### Execution Paths

**Need not be assessed** in unit testing. While execution paths might be relevant for more complex testing (e.g., integration or system testing), unit tests typically focus on isolated functions or methods, not the entire set of execution paths.

Option B is correct.

#### Q56 Text Solution:

In an **AVL tree**, rotations are performed to maintain balance when a new node is inserted, causing the tree to become unbalanced. Each type of rotation corresponds to a specific insertion scenario:

**LL Rotation:** The new node is inserted in the left sub-tree of the left sub-tree of the critical node. Correct because an LL imbalance occurs when a node is added to the left sub-tree of the left child, causing a left-heavy imbalance.

**RR Rotation:** The new node is inserted in the right sub-tree of the right sub-tree of the critical node.

Correct because an RR imbalance occurs when a node is added to the right sub-tree of the right child, causing a right-heavy imbalance.

**LR Rotation:** The new node is inserted in the right sub-tree of the left sub-tree of the critical node.

Correct because an LR imbalance occurs when a node is added to the right sub-tree of the left

child, requiring a left-right rotation to balance the tree.

**RL Rotation:** The new node is inserted in the left sub-tree of the right sub-tree of the critical node. Correct because an RL imbalance occurs when a node is added to the left sub-tree of the right child, requiring a right-left rotation to balance the tree.

#### Q57 Text Solution:

**Statement A: "Hamiltonian cycle problem is NP-Complete."**

**Correct.** The Hamiltonian cycle problem is an NP-complete problem, meaning it is both in NP and as hard as any problem in NP. This is the result of computational complexity theory.

**Statement B: "If G is an undirected Bipartite graph with an odd number of vertices, then G is Hamiltonian."**

**Incorrect.** A bipartite graph with an odd number of vertices cannot have a Hamiltonian cycle. In a bipartite graph, vertices are divided into two sets, and a Hamiltonian cycle requires that each vertex be visited exactly once, which would not be possible if there is an odd number of vertices.

**Statement C: "Hamiltonian path problem cannot be solved in polynomial time on directed acyclic graphs."**

**Incorrect**

The Hamiltonian path problem is NP-Complete meaning a proposed solution can be verified in polynomial time.

**Statement D: "If G is a connected undirected graph with at least three vertices and  $G^3$  is the graph obtained by connecting all pairs of vertices that are connected by a path in G of length at most 3."**

Correct. Based on graph theory, where the  $G^3$  notation indicates connections within a certain path length threshold. For any two vertices  $u$  and  $v$  in the original graph  $G$ , an edge is added between  $u$  and  $v$  in  $G^3$  if there exists a path of length 3 or less connecting them in  $G$ .

**Statement E: "If any NP-Complete problem is polynomial time solvable, then  $P=NP$ ."**



**Correct.** This is a foundational concept in computational complexity. If even one NP-complete problem can be solved in polynomial time, then all problems in NP can also be solved in polynomial time, which would imply  $P=NP$ .

**Q58 Text Solution:**

Statement A: "The running time is  $T(n)=\Theta(n^3)$ "

- **True:** In standard matrix multiplication using the divide and conquer approach, we divide an  $n \times n$  matrix into four  $n/2 \times n/2$  submatrices. This approach requires 8 recursive multiplications, and the running time typically ends up being  $\Theta(n^3)$ .

**Statement C: "The recurrence can be defined as  $T(n)=8T(n/2)+\Theta(n^2)$ "**

- **True:** This recurrence relation is correct for matrix multiplication with a divide and conquer approach. Here, we split the matrix into 8 subproblems (multiplications), each of size  $n/2$ . The additional  $\Theta(n^2)$  term accounts for the combining step.

**Statement E: "Master's theorem, one of the cases is applicable"**

- **True:** Master's theorem used to solve recurrence relations of the form  $T(n)=aT(n/b)+f(n)$ . For matrix multiplication with divide and conquer, the recurrence relation  $T(n)=8T(n/2)+\Theta(n^2)$  can be solved using Master's theorem.

Option A is correct.

**Q59 Text Solution:**

Option D is Correct

There are 4 Automata in the Table, analyzing each Automata one by one:

For Automata A,

- States:  $\{q_0\}$
- Start State:  $q_0$
- Accepting State:  $q_0$
- Transitions:
  - On 'a': stays at  $q_0$

- On 'b': stays at  $q_0$

Based on the provided information, as  $q_0$  is the only state and loops back to itself on both 'a' and 'b', it can process any string over  $\{a,b\}$ . In Being the accepting state, it can be concluded that it accepts all strings in L.

For Automata B,

- States:  $\{q_0, q_1\}$
- Start State:  $q_0$
- Accepting States:  $q_0, q_1$
- Transitions:
  - From  $q_0$ :
    - On 'a' or 'b': moves to both  $q_0$  and  $q_1$
  - From  $q_1$ :
    - On 'a' or 'b': stays at  $q_1$

Based on the provided information, it can be said that from the start state  $q_0$ , the automaton can always move to an accepting state. Hence, it can process any combination of 'a' and 'b' and end in an accepting state. Therefore, it accepts all strings in L.

For Automata C,

- States:  $\{q_0, q_1\}$
- Start State:  $q_0$
- Accepting States:  $q_0, q_1$
- Transitions:
  - From  $q_0$ :
    - On 'a': moves to  $q_0$  and  $q_1$
    - On 'b': stays at  $q_0$
  - From  $q_1$ :
    - On 'b': stays at  $q_1$
    - On 'a': no transition

Based on the provided information, it can be said that even though  $q_1$  has no transition on 'a',  $q_0$  can handle both 'a' and 'b'. The automaton can always process any string by staying in  $q_0$  or moving between  $q_0$  and  $q_1$  without getting stuck. Since both states are accepting, all strings in L are accepted.



For Automata D,

- States:  $\{q_0, q_1, q_2\}$
- Start State:  $q_0$
- Accepting State:  $q_0$
- Transitions:
  - From  $q_0$ :
    - On 'a' or 'b': stays at  $q_0$
  - From  $q_1$ :
    - On 'a': moves to  $q_2$
    - On 'b': no transition
  - From  $q_2$ :
    - On 'a': moves to  $q_0$
    - On 'b': stays at  $q_2$

Based on the information provided, it can be said that The start state  $q_0$  loops on both 'a' and 'b' and is the only accepting state. There are no transitions from  $q_0$  to  $q_1$  so  $q_1$  and  $q_2$  are unreachable. Effectively, the automaton behaves like Automaton A, accepting all strings in L by looping at  $q_0$ .

All four automata can process any string over  $\{a,b\}$  and accept it, meaning they all recognize the language  $L = \{a, b\}^*$ .

#### Q60 Text Solution:

Option A is Correct

Option A: Syntax Directed Translation can be applied in Type Checking.

It is Correct as SDT can indeed be used in type checking. Semantic rules attached to grammar productions can perform type checks on the attributes of syntactic constructs. For example, in a compiler, SDT can verify that an operation between variables respects type compatibility rules.

Option B: Syntax Directed Translation can be used in 3-address code generation.

It is Correct as SDT is commonly used in generating intermediate representations, such as 3-address code, by associating actions that produce the code with grammar productions. This is a key use of SDT in compiler construction.

Option C: Syntax Directed Translation can be used in the construction of a syntax tree.

It is Correct as SDT can be used to build syntax trees by associating actions with grammar productions that construct tree nodes. This is part of semantic analysis in compilers, where syntax trees represent the hierarchical structure of source code.

Option D: Syntax Directed Translation can be used to recognize valid tokens.

It is Incorrect as Token recognition is typically handled by the lexical analyzer (lexer), which operates before syntax analysis. SDT is applied during and after parsing (syntax analysis) and is not responsible for recognizing tokens. SDT focuses on attaching semantic actions to grammar productions rather than tokenizing input, which is outside its scope.

#### Q61 Text Solution:

**The router R has three neighboring routers X, Y, and Z with the following information:**

Distances from R to its neighbors:

R to X: 3

R to Y: 2

R to Z: 5

Routing vectors received by R from its neighbors for destination P:

X to P: 7

Y to P: 6

Z to P: 5

Routing vectors received by R from its neighbors for destination Q:

X to Q: 4

Y to Q: 6

Z to Q: 8

#### Calculations

Distance from R to P:

The distance from R to P through X is  $3+7=10$ .

The distance from R to P through Y is  $2+6=8$ .

The distance from R to P through Z is  $5+5=10$ .

Minimum distance to P is 8 via Y.

Distance from R to Q:

The distance from R to Q through X is  $3+4=7$ .

The distance from R to Q through Y is  $2+6=8$ .



The distance from R to Q through Z is  $5+8=13$ .

Minimum distance to Q is 7 via X.

**Explanation:**

Statement A: "The distance from R to P will be stored as 10."

This is incorrect because the minimum distance from R to P is 8, not 10.

Statement B: "The distance from R to Q will be stored as 7."

This is correct because the minimum distance from R to Q is indeed 7.

Statement C: "The next hop router for a packet from R to P is Y."

This is correct because the shortest path to P is through Y, with a distance of 8.

Statement D: "The next hop router for a packet from R to Q is Z."

This is incorrect because the shortest path to Q is through X, not Z.

Option A is correct.

**Q62 Text Solution:**

Option D is Correct

Analysing the Table

| Subnet Number | Subnet Mask         | Next Hop    |
|---------------|---------------------|-------------|
| 12.20.164.0   | 255.255.252.0 (/22) | R1          |
| 12.20.170.0   | 255.255.254.0 (/23) | R2          |
| 12.20.168.0   | 255.255.254.0 (/23) | Interface 0 |
| 12.20.166.0   | 255.255.254.0 (/23) | Interface 1 |
| Default       | -                   | R3          |

Each entry in the routing table represents a different subnet. The goal is to find a combination of CIDR blocks that fully cover the IP ranges of these subnets.

Step 2: Determine the IP Range for Each Subnet

1. 12.20.164.0/22 (Subnet Mask: 255.255.252.0)

The Range is 12.20.164.0 to 12.20.167.255

1. 12.20.170.0/23 (Subnet Mask: 255.255.254.0)

The Range is 12.20.170.0 to 12.20.171.255

1. 12.20.168.0/23 (Subnet Mask: 255.255.254.0)

The Range is 12.20.168.0 to 12.20.169.255

1. 12.20.166.0/23 (Subnet Mask: 255.255.254.0)

The Range is 12.20.166.0 to 12.20.167.255

Evaluate each CIDR option provided and determine which IP ranges they cover.

Option A: 12.20.164.0/20

CIDR Range for 12.20.164.0/20:

- This would cover IP addresses from 12.20.160.0 to 12.20.175.255.
- This option includes all addresses needed to cover the subnets in the routing table (12.20.164.0 to 12.20.171.255), but it also includes additional addresses (from 12.20.160.0 to 12.20.163.255 and from 12.20.172.0 to 12.20.175.255), which are not part of the routing table.

Hence, Option A does cover all required subnets but includes extra addresses outside the specified range, making it less precise.

Option B: 12.20.164.0/22

CIDR Range for 12.20.164.0/22:

- 12.20.164.0 in binary is:  
00001100.00010100.10100100.00000000
- /22 mask means the first 22 bits are fixed, allowing 2 bits for host addresses in the third octet.

Hence, This covers the IP range 12.20.164.0 to 12.20.167.255.

- IP Range Analysis:
  - 12.20.164.0/22 includes the IP addresses from 12.20.164.0 to 12.20.167.255.
  - This range covers only the first subnet (12.20.164.0/22).
  - However, it does not cover the following subnets:
    1. 12.20.168.0/23 (Range: 12.20.168.0 to 12.20.169.255)



2. 12.20.170.0/23 (Range: 12.20.170.0 to 12.20.171.255)

Hence, Option B (12.20.164.0/22) does not cover all subnets in the routing table. It only includes the range 12.20.164.0 to 12.20.167.255.

Option C: 12.20.164.0/21

CIDR Range for 12.20.164.0/21:

- 12.20.164.0 in binary is:  
00001100.00010100.10100100.00000000
- /21 mask means the first 21 bits are fixed, allowing 3 bits for host addresses in the third octet.
- This covers the IP range 12.20.160.0 to 12.20.167.255.
- IP Range Analysis:
  - 12.20.164.0/21 includes the IP addresses from 12.20.160.0 to 12.20.167.255.
  - This range only covers the first subnet 12.20.164.0 to 12.20.167.255, but it misses the remaining required subnets:
    1. 12.20.168.0/23 (Range: 12.20.168.0 to 12.20.169.255)
    2. 12.20.170.0/23 (Range: 12.20.170.0 to 12.20.171.255)

Hence, Option C does not cover all subnets in the routing table.

Option D: 12.20.168.0/22

CIDR Range for 12.20.168.0/22:

- 12.20.168.0 in binary is:  
00001100.00010100.10101000.00000000
- /22 mask means the first 22 bits are fixed, allowing 2 bits for host addresses in the third octet.
- This covers the IP range 12.20.168.0 to 12.20.171.255.
- IP Range Analysis:
  - 12.20.168.0/22 includes the IP addresses from 12.20.168.0 to 12.20.171.255.
  - This range covers only the following subnets:

1. 12.20.168.0/23 (Range: 12.20.168.0 to 12.20.169.255)

2. 12.20.170.0/23 (Range: 12.20.170.0 to 12.20.171.255)

- However, it does not cover the first subnet:

1. 12.20.164.0/22 (Range: 12.20.164.0 to 12.20.167.255)

Hence, Option D (12.20.168.0/22) does not cover all subnets in the routing table. It only includes the range 12.20.168.0 to 12.20.171.255, missing 12.20.164.0 to 12.20.167.255.

So, B and D individually do not cover all the subnets, combining them would provide full coverage:

- (12.20.164.0/22) covers 12.20.164.0 to 12.20.167.255.
- (12.20.168.0/22) covers 12.20.168.0 to 12.20.171.255.

Together, they cover the entire range 12.20.164.0 to 12.20.171.255 without any gaps and without including unnecessary addresses beyond this range.

### Q63 Text Solution:

#### MTU Values:

The maximum transfer unit (MTU) between P and R is 1500 bytes.

The MTU between R and Q is 820 bytes.

#### Packet:

A TCP segment of 1400 bytes is sent from P to Q. The IP header size is 20 bytes, so the total size of the IP datagram sent by P is: 1400 (TCP segment)+20 (IP header)=1420 bytes

#### Fragmentation:

Since the MTU between R and Q is 820 bytes, the 1420-byte packet will be fragmented at R. The 820 bytes include the IP header, so the maximum payload per fragment is: 820-20=800 bytes

#### Fragment Details:

##### First Fragment:

Payload: 800 bytes

IP header: 20 bytes



Total size: 820 bytes

Offset: 0

**Second Fragment:**

Remaining payload:  $1400 - 800 = 600$  bytes

IP header: 20 bytes

Total size: 620 bytes

Offset: 800 bytes

**Statement A: "Two fragments are created at R and IP datagram size carrying the second fragment is 620 bytes."**

True: Based on the calculation, two fragments are indeed created at R:

The first fragment is 820 bytes (800 bytes payload + 20 bytes header).

The second fragment is 620 bytes (600 bytes payload + 20 bytes header).

**Statement B: "If the second fragment is lost, then R will resend the fragment with IP Identification value of 0x1234."**

False: IP does not handle automatic retransmission of lost fragments. If a fragment is lost, the entire datagram (all fragments) must be resent by the original sender P. R does not resend fragments on its own.

**Statement C: "If the second fragment is lost, then P is required to resend the entire TCP segment."**

True: Since IP does not provide for individual fragment retransmission, the entire TCP segment must be resent by P if any fragment is lost.

**Statement D: "TCP destination port can be determined by analyzing the second fragment only."**

False: The TCP header, which contains the destination port, is located in the first fragment. The second fragment does not contain this information, so analyzing only the second fragment will not reveal the TCP destination port. Option B is correct.

**Q64 Text Solution:**

**Statement A: "It is essentially a machine learning algorithm."**

Correct. Neural networks are a type of machine learning algorithm inspired by the structure and

function of the human brain. They are widely used in supervised, unsupervised, and reinforcement learning tasks.

**Statement B: "It is useful when solving the problems for which the data set is very large."**

Correct. Neural networks, especially deep neural networks, excel in processing large datasets and are particularly effective in applications such as image recognition, natural language processing, and other areas that require large volumes of data.

**Statement C: "They are able to extract features without input from the programmer."**

Correct. Neural networks, especially in deep learning, can automatically learn features from data during the training process. This is known as feature extraction, where the network identifies patterns and hierarchies in the data without explicit programming.

**Statement D: "These are systems modeled on the human brain and nervous system."**

Correct. Neural networks are inspired by the structure of the human brain, with layers of neurons (nodes) that simulate the brain's ability to recognize patterns and make decisions.

The correct option is D.

**Q65 Text Solution:**

The **rule-based approach** in artificial intelligence and machine learning has several limitations:

**Statement A: "The learning is static."**

Correct. In a rule-based system, the knowledge is predefined, and there is no mechanism for the model to learn or adapt. Once the rules are set, the system operates according to those fixed rules, making the learning process static.

**Statement B: "Any changes made to the original data will not be considered."**

Correct. Rule-based systems do not dynamically adapt to new data or changes in existing data. The system will continue to apply the same rules, regardless of any updates or changes to the underlying data.

**Statement C: "Once trained, the model cannot improve on the basis of feedback."**



Correct. Rule-based models do not have a mechanism to improve based on feedback. They lack the ability to adjust rules or learn from new information, as they do not involve training in the way that machine learning models do.

The correct option is A.

**Q66 Text Solution:**

| List I                           | List II                | Explanation                                                            |
|----------------------------------|------------------------|------------------------------------------------------------------------|
| A. Linear Program                | III. Simplex Algorithm | The Simplex Algorithm is used to solve linear programming problems.    |
| B. Predicate Logic               | IV. First order logic  | Predicate Logic is also known as First Order Logic.                    |
| C. Tree Traversal                | II. In order           | "In order" is one of the methods for tree traversal.                   |
| D. Minimum Spanning Tree Problem | I. Prim's Algorithm    | Prim's Algorithm is used to find the Minimum Spanning Tree in a graph. |

The correct option is B.

**Q67 Text Solution:**

The correct answer is **option B**.

| List I | List II                        |
|--------|--------------------------------|
| A. DAA | IV. Arithmetic Instruction     |
| B. LXI | III. Data Movement Instruction |
| C. RST | II. Interrupt Instruction      |
| D. JMP | I. Program Control Instruction |

- **DAA (Decimal Adjust for Addition):** This instruction is used in assembly language to

adjust the result of an addition operation to represent it in BCD (Binary-Coded Decimal) format. It modifies the contents of the accumulator based on the last addition operation.

- **LXI (Load Register Pair Immediate):** This instruction loads a register pair with a specified 16-bit immediate value. It is primarily used for initializing registers.
- **RST (Restart):** The RST instruction calls a specific interrupt service routine by resetting the program counter to a predefined address. This is categorized as an Interrupt instruction, as it triggers an interrupt and alters the normal flow of execution.
- **JMP (Jump):** The JMP instruction causes the program to jump to another part of the code, effectively altering the control flow. This is classified as a Program Control Instruction since it determines where execution continues in the program.

**Q68 Text Solution:**

The correct answer is **option B ( A-III, B-I, C-IV, D-II )**.

| List I                 | List II                   |
|------------------------|---------------------------|
| A. 2-D Array Input     | III. Camera               |
| B. Transmissive Device | I. Liquid Crystal Display |
| C. Binary Image        | IV. Inkjet printer        |
| D. 1-D Array Input     | II. Flat-bed scanner      |

**A. 2-D Array Input:** A 2-D array input typically refers to a data structure that can hold data in two dimensions, often used in image processing or graphical representations. Devices that utilize this include cameras, which capture images as arrays of pixels.

**B. Transmissive Device:** A transmissive device is one that allows light to pass through it. Common examples include LCDs (Liquid Crystal Displays), which use liquid crystals to modulate light for display purposes.

**C. Binary Image:** A binary image consists of



pixels that can take on one of two values, usually representing black and white (0 and 1). This type of image is commonly used in digital imaging and processing applications. An inkjet printer produces images by spraying tiny droplets of ink onto paper, typically using binary image data for printing.

**D. 1-D Array Input:** A 1-D array input refers to a linear data structure, often used for simpler forms of data representation or input devices like keyboards or simple sensors.

**Q69 Text Solution:**

The correct answer is **option B ( A-III, B-IV, C-I, D-II )**.

**A. Timestamp protocol:** The timestamp protocol is a concurrency control mechanism that assigns a unique timestamp to each transaction. The main feature of this protocol is that it ensures transactions are executed based on their timestamps, which helps maintain consistency without locking resources. If a transaction attempts to access data that has been modified by another transaction with an earlier timestamp, it will be rolled back or delayed. This approach also prevents deadlock situations.

**B. Strict two-phase locking protocol:** Strict two-phase locking (S2PL) is a locking protocol where all locks are acquired before any locks are released. This guarantees serializability and prevents cascading rollbacks but can lead to deadlocks if not managed properly.

**C. Two-phase locking:** Two-phase locking (2PL) is a concurrency control method that requires transactions to acquire all necessary locks before releasing any. It consists of two phases: the growing phase (where locks are acquired) and the shrinking phase (where locks are released). While it ensures serializability, it does not prevent deadlocks.

**D. Lock point in two-phase locking protocol:**

The lock point refers to the moment in time when a transaction acquires its last lock during its execution. In 2PL, this point is crucial because it

determines when other transactions can see changes made by this transaction.

**Q70 Text Solution:**

The correct answer is **option D( A-III, B-I, C-IV, D-II )**.

Reference Bit (R) indicates whether a page has been accessed (read or written) recently. A value of 0 means it has not been accessed, while a value of 1 means it has been accessed.

Modify Bit (M) indicates whether a page has been modified (written to) since it was loaded into memory. A value of 0 means it has not been modified, while a value of 1 means it has been modified.

| List I                             | List II                                                     |
|------------------------------------|-------------------------------------------------------------|
| A. (reference bit=0, modify bit=0) | III. Best page to replace.                                  |
| B. (reference bit=0, modify bit=1) | I. Must write out before replacement.                       |
| C. (reference bit=1, modify bit=0) | IV. Probably will be used again soon.                       |
| D. (reference bit=1, modify bit=1) | II. Probably will be used again soon and need to write out. |

**Q71 Text Solution:**

The correct answer is **option B ( A-III, B-I, C-IV, D-II )**.

**Condition Coverage (A):** This is a white box testing technique that ensures that each condition in a decision statement has been executed both to true and false. It is primarily used in unit testing to verify the logic of the code.

**Equivalence Class Partitioning (B):** This is a black box testing method that divides input data into partitions of equivalent data from which test cases can be derived. The idea is that if one condition in a partition works, all conditions should work.

**Volume Testing (C):** This type of testing is related to performance testing where the system



is subjected to a large volume of data to evaluate its behavior under load. It checks how well the system handles high volumes of data.

**Alpha Testing (D):** It is a type of system testing that's performed during the software development process and is a form of acceptance testing performed by internal staff before releasing the product to external users. It focuses on finding bugs before the software goes live.

**Q72 Text Solution:**

The correct answer is **option C ( A-IV, B-I, C-II, D-III )**.

**A. 8-Queens Problem:** The 8-Queens problem is a problem where the goal is to place eight queens on an 8x8 chessboard so that no two queens threaten each other. This problem is solved using backtracking.

**B. Closest Pair of Points:** The closest pair of points problem involves finding the two points that are closest together in a set of points in a plane. This can be efficiently solved using a divide-and-conquer approach.

**C. String Edit Problem:** The string edit problem measures how dissimilar two strings are by counting the minimum number of operations required to transform one string into another. This is solved using dynamic programming.

**D. Knapsack Problem:** The problem involves selecting items with given weights and values to maximize value without exceeding weight capacity. There are various approaches to solving this problem, but it can be effectively approached using a greedy method for certain variations (like fractional knapsack) or dynamic programming for others (like 0/1 knapsack). However, for this context, we will associate it with greedy methods due to its frequent application in simpler forms.

**Q73 Text Solution:**

The correct answer is **option B**.

**LL Parsing:**

- The initial configuration generally has the stack containing the start symbol (like 'S') and the input string (like 'w').
- The final configuration typically has the stack empty and the input string fully consumed.

**LR Parsing:**

- The initial configuration for LR parsing starts with the stack empty, and the input contains the string to be parsed. This corresponds to  $w\$$ , where  $w$  is the input string, and the stack is empty.
- In the final configuration for LR parsing, the stack will contain the start symbol (denoted as  $S$ ), and the input will be empty once the parsing process is complete. This corresponds to  $\$ S \$$ , where the stack contains  $S$ , and the input is empty.

**Q74 Text Solution:**

The correct answer is **option C ( A-III, B-I, C-II, D-IV )**.

**Physical Layer (A):** The Physical Layer is responsible for the transmission of raw bits over a communication channel. This includes defining the electrical and physical specifications of the data connection.

**Data Link Layer (B):** The Data Link Layer is responsible for transforming raw bits into frames for transmission. It provides node-to-node data transfer and handles error correction from the Physical Layer.

**Network Layer (C):** The Network Layer is responsible for controlling and monitoring subnet traffic and routing packets across different networks. It manages addressing and routing through datagrams.

**Transport Layer (D):** The Transport Layer is responsible for ensuring complete data transfer through either connection-oriented or



connection-less services using segments or datagrams.

**Q75 Text Solution:**

The correct answer is **option B ( A-II, B-II, C-I, D-II )**.

**A. Regression:** This is a statistical method used for predicting a continuous outcome variable based on one or more predictor variables. It falls under the category of supervised learning because it requires labeled data to train the model.

**B. KNN (K-Nearest Neighbors):** This is a classification algorithm that assigns a class to a sample based on the classes of its nearest neighbors. KNN is also classified as supervised learning since it uses labeled data for training.

**C. Clustering:** This refers to grouping a set of objects in such a way that objects in the same group (or cluster) are more similar than those in other groups. Clustering is an unsupervised learning technique because it does not require labeled data.

**D. Decision Tree:** This is another supervised learning algorithm used for both classification and regression tasks. It creates a model that predicts the value of a target variable by learning simple decision rules inferred from the data features.

**Q76 Text Solution:**

The correct answer is **option D**.

A connected component in a graph is a subset of the graph where there is a path between any two vertices within that subset, and no vertex in the subset is connected to any vertex outside it.

**Step 1: Minimum Edges for Each Component**

For each connected component, at least one edge is required to connect two vertices.

However, if we want to minimize the number of edges while still maintaining connectivity within each component, we can use trees as our structure. A tree is a type of graph that is minimally connected; it has exactly  $v-1$  edges for  $v$  vertices. Thus, for each connected component containing  $n_i$  vertices (where  $i = 1, 2, \dots, k$ ),

The minimum number of edges needed to keep that component connected is:  $n_i - 1$

**Step 2: Total Vertices and Components**

Since we have  $n$  total vertices distributed across  $k$  components, we can express this as:

$$n_1 + n_2 + \dots + n_k = n$$

**Step 3: Calculating Minimum Edges for All Components**

The total minimum number of edges required for all  $k$  components would be:

$$(n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1)$$

$$= (n_1 + n_2 + \dots + n_k) - k = n - k$$

Therefore, the minimum possible number of edges in a simple undirected graph with  $n$  vertices and  $k$  connected components are  $n - k$

**Q77 Text Solution:**

The correct answer is **option B**.

1. **(D) Program Counter (PC)** holds the address of the next instruction that needs to be fetched from memory. It is updated after each instruction fetch.
2. **(C) Memory Address Register (MAR)** holds the address of the memory location from which data or an instruction is to be fetched. During a fetch operation, the address stored in the PC is transferred to the MAR.
3. **(A) Memory** is where instructions and data are stored. Once the MAR has the address, it accesses memory to retrieve the instruction.
4. **(B) Memory Buffer Register (MBR)** is also known as the Memory Data Register (MDR), this register temporarily holds data that has been read from or written to memory. After fetching, the instruction retrieved from memory is placed into the MBR.
5. **(E) Instruction Register (IR)** holds the actual instruction that has been fetched from memory and is ready for decoding and execution.

**Q78 Text Solution:**

The correct answer is (D, B, A, C).

**(A) Merge Sort**  $\Rightarrow$  Worst-case Complexity:  $O(n \log n)$



Merge sort is a divide-and-conquer algorithm that splits the array into halves, recursively sorts them, and then merges the sorted halves. The logarithmic factor arises from the division of the array, while the linear factor comes from merging.

**(B) Linear Search**  $\Rightarrow$  Worst-case Complexity:

$O(n)$

Linear search scans through each element of an array sequentially until it finds the target value or reaches the end of the array. In the worst case, it checks every element once.

**(C) Bubble Sort**  $\Rightarrow$  Worst-case Complexity:  $O(n^2)$

Bubble sort repeatedly steps through the list, compares adjacent elements, and swaps them if they are in the wrong order. This process is repeated until no swaps are needed. In the worst case (when the array is sorted in reverse order),  $n$  is required to pass through  $n$  elements.

**(D) Binary Search**  $\Rightarrow$  Worst-case Complexity:

$O(\log n)$

Binary search works on sorted arrays by repeatedly dividing the search interval in half. It eliminates half of the remaining elements with each comparison, leading to logarithmic time complexity.

The ascending order of search and sort algorithms of their worst-case complexities is (D, B, A, C).

**Q79 Text Solution:**

The correct answer is **option B**.

**The Schedule S**

At Timestamp t1:

T2 reads X (R(X)) & T3 reads Y (R(Y)).

At Timestamp t2:

T2 writes to X (W(X)) & T3 writes to Y (W(Y)).

At Timestamp t3:

T1 reads Y (R(Y)), which T3 wrote at t2.

T4 reads X (R(X)), which T2 wrote at t2.

At Timestamp t4:

T1 writes Y (W1(Y)) & T4 writes X (W4(X)).

At Timestamp t5:

T3 reads X (R3(X)), which T2 wrote at t2 & T3 writes to X (W3(X)).

### Determining Dependencies

From the above analysis, we can deduce the following dependencies:

T2 must execute before T3 because T3 reads Y after T2 has written it.

T2 must execute before T4 because T4 reads X after T2 has written it.

T3 must execute before T1 because T1 reads Y after T3 has written it.

T1 must execute before T4 because T4 writes X after T1 has read Y.

### Execution Order

Given the dependencies, we can deduce the following execution order:

T2 must execute first (it writes X).

T3 must execute next (it reads Y and writes X).

T1 must execute after T3 (it reads Y).

T4 can execute last (it reads X and writes X).

Thus, the correct execution order that respects these dependencies is: T2-T3-T1-T4

**Q80 Text Solution:**

The correct answer is **option A**.

**The C\_LOOK algorithm** services the requests in a way that the head moves in one direction (in this case, towards the higher end) and only services requests that are ahead of the current position of the head. It wraps around the beginning of the queue only after reaching the end of the available requests.

**Given:**

The initial position of the head: 101

Read request sequence: 123, 45, 197, 87, 250, 198, 280

The head is moving towards the higher end of the disk.

**Step 1: Sort the Requests**

First, we need to sort the requests that are



greater than or equal to the current position of the head (101):

Requests greater than or equal to 101: 123, 197, 198, 250, 280

Requests less than 101: 45, 87

### Step 2: Serve the Requests

The head will first serve the requests in the order from the current position (101) to the highest request (280). After reaching the highest request (280), it will jump back to the lowest request that is still pending (45, 87). Order of Serving Requests,

Start at 101

Move to 123 (head movement =  $123 - 101 = 22$ )

Move to 197 (head movement =  $197 - 123 = 74$ )

Move to 198 (head movement =  $198 - 197 = 1$ )

Move to 250 (head movement =  $250 - 198 = 52$ )

Move to 280 (head movement =  $280 - 250 = 30$ )

Now the head has reached the highest request (280).

### Step 3: Jump Back to the Lowest Pending Request

After serving all requests above 101, the head will jump back to the lowest pending request (45).

Jump back to 45 (head movement =  $280 - 45 = 235$ )

Move to 87 (head movement =  $87 - 45 = 42$ )

**Total Head Movement Calculation** = sum up all the head movements

From 101 to 123: 22

From 123 to 197: 74

From 197 to 198: 1

From 198 to 250: 52

From 250 to 280: 30

From 280 to 45: 235

From 45 to 87: 42

Total head movement =  $22 + 74 + 1 + 52 + 30 + 235 + 42 = 456$

### Q81 Text Solution:

The correct answer is **option B**.

### The incremental software development model

involves developing a system in small, manageable segments or increments. Each increment builds upon the previous ones, adding new features and functionality over time. It is a good approach when a working core product is required quickly.

One of the main advantages of this approach is its ability to quickly deliver functional components of a product. By concentrating on developing a core set of features first, teams can release a minimum viable product (MVP) sooner than they could with other methodologies. This early release allows stakeholders to provide feedback throughout the process, which can then guide the development of subsequent increments.

### Q82 Text Solution:

Explanation:

Visit Node A

(A → C, Distance = 2, and A → F Distance = 7, Compare  $\min(2, 7) = 2$ , Node A → C)

Visit Node C

(C → B, Distance = 10, and C → F Distance = 3, Compare  $\min(10, 3) = 3$ , Node C → F)

Visit Node F

(F → D, Distance = 1, and F → E Distance = 6, Compare  $\min(1, 6) = 1$ , Node F → D)

Visit Node D

(D → Z, Distance = 7, and D → E Distance = 2, Compare  $\min(7, 2) = 2$ , Node D → E)

Visit Node E

(E → Z, Distance = 4, Node E → Z)

**The Final Answer is: A → C → F → D → E**

### Q83 Text Solution:

The correct answer is **option C**.

The precedence of regular expression operators from lowest to highest is union, concatenation, and star.

Union has the lowest precedence, concatenation has the next highest precedence and star has the highest precedence.

**For example**, the regular expression  $a + b * c$



would be grouped as  $a + ((b^*)c)$ .

**Q84 Text Solution:**

The correct answer is **option C**.

To generate keys for the RSA algorithm, the correct order of operations is as follows:

D: Choose two large prime numbers ( $p$ ) and ( $q$ ).

B: Compute key length ( $n = p \text{ times } q$ ).

E: Compute ( $\lambda(n)$ ) (the Carmichael function or Euler's totient function).

A: Choose an integer ( $e$ ) such that ( $1 < e < \lambda(n)$ ) and ( $e$ ) is coprime to ( $\lambda(n)$ ).

C: Find ( $d$ ) which is the modular inverse of ( $e$ ) modulo ( $\lambda(n)$ ), i.e., ( $d$  equivalent  $e^{-1} \bmod \lambda(n)$ ).

Based on this sequence, the correct order is: D-B-E-A-C

**Q85 Text Solution:**

The correct answer is **option A**.

**A: Information → Input layer → Hidden layer**

This sequence is correct. In a neural network, information (data) is fed into the input layer. The input layer then passes this information to one or more hidden layers, where computations and transformations occur to learn patterns in the data.

**Q86 Text Solution:**

The correct answer is **option A**.

**Statement I is True:** The main objective of using virtual memory is to increase the effective capacity of the memory system.

Virtual memory is a memory management feature that helps a computer deal with shortages of physical memory by temporarily moving data from random access memory (RAM) to disk storage. This process allows a system to run larger applications or multiple applications at the same time than it could with only the physical RAM available.

**Statement II is true:** The size of virtual memory can be larger than the main memory.

Virtual memory can be larger than physical RAM. This is because virtual memory uses disk space (such as hard drives or SSDs) to extend available

addressable space beyond what is physically present in RAM.

Statement I correctly identifies one of the primary purposes of virtual memory and Statement II accurately reflects that virtual memory can exceed the size of physical RAM.

**Q87 Text Solution:**

The correct answer is **option A**.

**Statement I is true:** Any retrieval request that is specified in the basic relational algebra can also be specified in relational calculus.

Relational algebra is a procedural query language that provides a set of operations to manipulate relations (tables). It allows users to construct queries step-by-step. Relational calculus, on the other hand, is a non-procedural query language that specifies what data to retrieve without detailing how to retrieve it. There are two forms of relational calculus: tuple relational calculus (TRC) and domain relational calculus (DRC).

The expressive power of both languages is equivalent when it comes to retrieving data from a database.

**Statement II is true:** Any retrieval request that is specified in the relational calculus can also be specified in basic relational algebra.

Both languages have equivalent expressive power. This means that any retrieval request formulated in relational calculus can be translated into an equivalent expression using basic relational algebra operations.

**Q88 Text Solution:**

The correct answer is **option A**.

**Statement A is true:** Solicit stakeholder needs and expectations.

This is a fundamental objective of any software development process. Understanding what stakeholders need and expect is crucial for ensuring that the final product meets their requirements.



**Statement B is true:** Specify the software requirements for the product and post-development processes.

Specifying software requirements is an important task in software engineering, as it states what needs to be developed and how it should function after deployment.

**Statement C is true:** Generate the integrated technical plans for accomplishing the software development effort.

Creating integrated technical plans is essential for coordinating various aspects of software development, including timelines, resources, and methodologies.

**Statement D is true:** Translate stakeholder requirements into a complete, specified architectural description.

This statement reflects an important aspect of software architecture where stakeholder requirements are transformed into a structured design that guides development.

**Statement E is true:** Ensure that the software technical data package is sufficiently detailed to facilitate an efficient and effective software implementation.

Having a detailed technical data package is vital for ensuring that all team members understand how to implement the software effectively.

#### Q89 Text Solution:

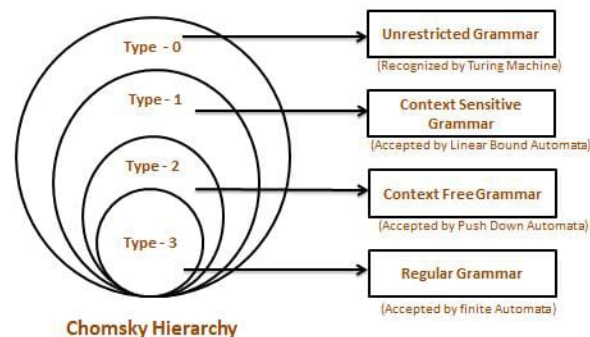
The correct answer is **option D**.

**Statement A is true:** For every regular language, we can design a Turing Machine.

Regular languages are a subset of formal languages that can be recognized by finite automata. Since Turing Machines are more powerful than finite automata, they can recognize all regular languages.

**Statement B is true:** For every context-free language, we can design a Turing Machine.

Context-free languages (CFLs) are also a subset of formal languages that can be recognized by pushdown automata. Turing Machines can recognize all context-free languages as well since they have greater computational power than pushdown automata.



#### Q90 Text Solution:

The correct answer is **option B**.

**Assertion A** states that Artificial Neural Networks (ANNs) are the backbone of deep learning algorithms, which is true. ANNs are a fundamental component of deep learning, as they are used to build complex models that can learn and represent data.

**Reason R** states that backpropagation is one of the techniques used to train an ANN, which is also true. Backpropagation is a widely used algorithm for training ANNs, as it allows the network to learn from its errors and adjust its weights and biases to improve its performance.

However, R is not the correct explanation of A. The fact that backpropagation is used to train ANNs does not explain why ANNs are the backbone of deep learning algorithms. The correct explanation would involve the ability of ANNs to learn complex patterns in data, their scalability, and their ability to be composed into deeper architectures, among other factors.

#### Q91. Text Solution:

The correct answer is **option B**.

To solve the fractional knapsack problem, we need to determine how to maximize the profit while adhering to a maximum weight capacity of



15. The approach involves calculating the profit-to-weight ratio for each item, sorting them based on this ratio, and then selecting items until we reach the maximum capacity.

**Step 1: Calculate Profit-to-Weight Ratios**

| Item no. | Profit (P) | Weight (W) | P/W ( Ratio ) |
|----------|------------|------------|---------------|
| 1        | 9          | 2          | 4.5           |
| 2        | 5          | 5          | 1             |
| 3        | 2          | 6          | 0.33          |
| 4        | 7          | 11         | 0.63          |
| 5        | 6          | 1          | 6             |
| 6        | 16         | 9          | 1.77          |
| 7        | 9          | 3          | 3             |

**Step 2: Sort Items by Profit-to-Weight Ratio**

| Item no | Ratio (P/W) | Weight (W) | Profit (P) |
|---------|-------------|------------|------------|
| 5       | 6           | 1          | 6          |
| 1       | 4.5         | 2          | 9          |
| 7       | 3           | 3          | 9          |
| 6       | 1.77        | 9          | 16         |
| 2       | 1           | 5          | 5          |
| 4       | 0.63        | 11         | 7          |
| 3       | 0.33        | 6          | 2          |

**Step 3:** Now, we start filling the knapsack with the items in the sorted order until the maximum capacity of 15 is reached:

Item 5: weight = 1, profit = 6 (total weight = 1, total profit = 6)

Item 1: weight = 2, profit = 9 (total weight = 3, total profit = 15)

Item 7: weight = 3, profit = 9 (total weight = 6, total profit = 24)

Item 6: weight = 9, profit = 16 (total weight = 15, total profit = 40)

Since the total weight has reached the maximum capacity of 15, we stop here. The optimal solution for the fractional knapsack problem is to include items 5, 6, 1, and 7, with a total profit of 40.

**Q92. Text Solution:**

The correct answer is **option A**.

To solve the fractional knapsack problem, we need to determine how to maximize the profit while adhering to a maximum weight capacity of 25. The approach involves calculating the profit-to-weight ratio for each item, sorting them based on this ratio, and then selecting items until we reach the maximum capacity.

**Step 1: Calculate Profit-to-Weight Ratios**

| Item no. | Profit (P) | Weight (W) | P/W (Ratio) |
|----------|------------|------------|-------------|
| 1        | 9          | 2          | 4.5         |
| 2        | 5          | 5          | 1           |
| 3        | 2          | 6          | 0.33        |
| 4        | 7          | 11         | 0.63        |
| 5        | 6          | 1          | 6           |
| 6        | 16         | 9          | 1.77        |
| 7        | 9          | 3          | 3           |

**Step 2: Sort Items by Profit-to-Weight Ratio**

| Item no. | Ratio (P/W) | Weight (W) | Profit |
|----------|-------------|------------|--------|
| 5        | 6           | 1          | 6      |
| 1        | 4.5         | 2          | 9      |
| 7        | 3           | 3          | 9      |
| 6        | 1.77        | 9          | 16     |
| 2        | 1           | 5          | 5      |
| 4        | 0.63        | 11         | 7      |
| 3        | 0.33        | 6          | 2      |

**Step 3:** Now, we start filling the knapsack with the items in the sorted order until the maximum capacity of 25 is reached:

Item 5: weight = 1, profit = 6 (total weight = 1, total profit = 6)

Item 1: weight = 2, profit = 9 (total weight = 3, total profit = 15)

Item 7: weight = 3, profit = 9 (total weight = 6, total profit = 24)

Item 6: weight = 9, profit = 16 (total weight = 15, total profit = 40)

Item 2: weight = 5, profit = 5 (total weight = 20, total profit = 45)

Item 4: weight = 11, profit = 7 (total weight = 31,



total profit = 52)

Since the total weight has exceeded the maximum capacity of 25, we need to remove the last item (Item 4) and take a fraction of it to fill the remaining capacity.

The remaining capacity is  $25 - 20 = 5$ . We can take a fraction of Item 4, which has a weight of 11. The fraction we can take is  $5/11$ , which gives us a profit of  $(5/11) * 7 = 3.18$ .

Therefore, the optimal solution for the fractional knapsack problem is to include items 5, 6, 1, 7, 2, and a fraction of Item 4, with a total profit of  $45 + 3.18 = 48.18$ .

**Q93. Text Solution:**

The correct answer is **option C**.

To solve the fractional knapsack problem, we use a greedy algorithm that involves sorting the items based on their value-to-weight ratio. The time complexity of this algorithm is determined by the sorting step, which is  $O(n \log n)$ , where  $(n)$  is the number of items.

In this case, we have a total of 8 items (the original 7 plus the new item). Therefore, the complexity for sorting these items would be:  $[O(8 \log 8)]$

Calculating  $(8 \log 8)$ :

$(\log 8 = 3)$  (since  $(8 = 2^3)$ )

Therefore,  $O(8 \log 8) = O(8 \text{ times } 3) = O(24)$ .

The best-case time of the fractional knapsack for this case is  $O(24)$ .

**Q94. Text Solution:**

The correct answer is **option C**.

The fractional knapsack problem is best solved using a Greedy Algorithm. This is because the greedy approach allows you to select the item with the highest value-to-weight ratio first and take fractions of items if necessary, which is the core principle of the fractional knapsack problem.

In contrast, the other options listed (Divide and Conquer, Dynamic Programming, and Backtracking) are more suited for solving the 0/1 knapsack problem, where items cannot be broken into smaller parts.

Therefore, the most efficient solution for the fractional knapsack problem with maximum capacity 15 can be obtained by using greedy algorithm.

**Q95. Text Solution:**

The correct answer is **option D**.

Option A is false: "0/1 knapsack can be applied for all kinds of problems but fractional knapsack can be applied only if the problem meets certain conditions." This statement is incorrect because both types of knapsacks are applicable under different scenarios. The 0/1 knapsack does not apply universally to all problems as it has specific constraints (items cannot be divided), while the fractional knapsack applies when items can be divided.

Option B is false: "There is no difference between the two. They give the same solution for any problem."

This statement is incorrect. The two problems have fundamentally different approaches and solutions due to their constraints on item selection (whole vs. fractional).

Option C is false: "We use the greedy algorithm for 0 / 1 knapsack and dynamic programming for fractional knapsack." This statement is also incorrect. The greedy algorithm is typically used for solving the fractional knapsack problem effectively, while dynamic programming is employed for solving the 0/1 knapsack problem due to its combinatorial nature.

Option D is true: "In 0/1 knapsack problem, items cannot be divided into smaller portions, and in fractional knapsack, we can divide the item into required proportions of weight." This statement accurately describes both types of problems. It correctly states that in a 0/1 knapsack, items must be taken whole or left out entirely, whereas, in a fractional knapsack, items can be divided into smaller portions.



**Q96. Text Solution:**

The correct answer is **option B**.

**Correct : (B)** Select s.rollno, s.name From student s, enrollment e Where e.rollno =s.rollno and e.courseid = ALL (select precourseid from prerequisite p where p.courseid = "324")

- This query selects students who have enrolled in all prerequisite courses for course 324.
- The use of ALL ensures that the student must have taken every course listed in the subquery (i.e., all prerequisite courses).
- This query is correct because it ensures that the student has enrolled in all prerequisite courses.

**Q97. Text Solution:**

The correct answer is **option A**.

Given Query:

SELECT s.rollno

FROM student s

WHERE s.sex = ALL (SELECT f.sex FROM professor f WHERE f.empid = s.advisor)

This query retrieves the roll numbers of all students whose gender matches the gender of their advisor.

**True: S1** It has a correlated subquery.

The subquery (SELECT f.sex FROM professor f WHERE f.empid = s.advisor) is correlated because it references the outer query's s.advisor. The subquery depends on the value of s.advisor from the outer query, making it correlated.

**False: S2** It has an uncorrelated subquery.

The subquery is correlated because it relies on the outer query's s.advisor. An uncorrelated subquery would not reference any columns from the outer query.

**True: S3** If we replace ALL with ANY in the above query, its result will be different.

The ALL operator requires that the condition hold true for all values returned by the subquery. In contrast, ANY would return true if the condition holds for at least one value returned by

the subquery. Therefore, changing ALL to ANY could change the query's results, especially if there are multiple advisors with different genders.

**False: S4** If we replace ALL with ANY in the above query, its result will be the same.

As explained in S3, using ANY instead of ALL can lead to different results. ANY checks if the condition is true for at least one value, while ALL checks it for all values.

**Q98. Text Solution:**

The correct answer is **option A**.

**Schema:**

department (deptid, name, hod, phone): Contains department details.

course (courseid, cname, credits, deptno):

Contains course details, including the department (deptno) it belongs to.

**Correct : A** SELECT deptid, name, SUM(credits) AS totalcredits

FROM department, course

WHERE deptid = deptno

GROUP BY deptid, name;

This query selects the deptid and name from the department table and computes the total credits for courses offered by each department. It uses a join condition deptid = deptno to link the department and course tables. The GROUP BY clause groups the results by deptid and name, allowing the SUM(credits) function to aggregate the total credits for each department.

**Q99. Text Solution:**

The correct answer is option D.

**Correct : D** SELECT deptid, name, SUM(credits) AS totalcredits

FROM department, course

WHERE deptid = deptno

GROUP BY deptid, name

HAVING totalcredits > 40;

This query correctly selects the department ID and name, calculates the total credits for courses offered by the department, and uses the WHERE clause to filter the courses based on their



department.

It groups the results by deptid and name, and then the HAVING clause is used to filter departments where the total credits are greater than 40.

**Q100. Text Solution:**

The correct answer is **option C**.

Given Query:

```
SELECT s.rollno, s.name
FROM student s
WHERE EXISTS (
 SELECT c.courseid
 FROM course c
 WHERE c.deptno = 5
 AND EXISTS (
 SELECT e.*
 FROM enrollment e
 WHERE e.courseid = c.courseid
 AND e.rollno = s.rollno
)
);
```

Outer Query:

The outer query selects the rollno and name of students from the student table.

First EXISTS Clause:

The outer query includes a condition that checks for the existence of at least one course in the course table that belongs to department number 5 (c.deptno = 5).

Inner EXISTS Clause:

For each course found in the first EXISTS clause, it checks if there exists an enrollment record in the enrollment table where the rollno of the student matches (e.rollno = s.rollno) and the courseid matches the course being checked (e.courseid = c.courseid).

This means that the student must be enrolled in at least one course from department number 5.

\* The query retrieves the roll number and names of students who are enrolled in at least one course offered by department number 5.



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